Journal of Econometrics 244 (2024) 105850

Contents lists available at ScienceDirect

Journal of Econometrics

journal homepage: www.elsevier.com/locate/jeconom

Large Bayesian SVARs with linear restrictions☆

Chenghan Hou

Hunan University, China

ARTICLE INFO

JEL classification:
C11
C32
C55
E52

Keywords:
Large vector autoregression
Equality and inequality restrictions
Over-identifying restrictions
Proxy-SVAR

ABSTRACT

This paper develops a Markov Chain Monte Carlo (MCMC) algorithm for Bayesian inference in large structural vector autoregressions (SVARs) with linear restrictions. Our proposed method is based on a novel parameter transformation scheme, which aims to facilitate sampling from the posterior distribution of model parameters when linear equality and inequality restrictions are imposed on contemporaneous impulse responses. A prominent feature of the proposed methodology is its applicability for inference in SVARs with over-identifying restrictions. In our empirical application, we demonstrate the usefulness of our method by employing a large Proxy-SVAR with multiple proxy variables to simultaneously identify multiple macroeconomic shocks and investigate their contributions to the 2007–09 Recession.

1. Introduction

Since the seminal work by Sims (1980), structural vector autoregressive models (SVARs) have become one of the most popular tools in empirical macroeconomics. Despite their popularity, obtaining convincingly identified structural shocks remains a major issue in SVAR analysis. In early SVAR studies, a recursive identification scheme was commonly used for identifying structural shocks. Although recursive identification is easy to implement and computationally convenient, it is too restrictive as it precludes many possible patterns of contemporaneous responses of endogenous variables to structural shocks, making it unsuitable for a broad range of empirical applications. One attractive alternative approach in the literature for achieving flexible and credible identification for SVARs is through the use of economically meaningful equality and inequality restrictions. The main objective of this paper is to develop an algorithm for Bayesian inference in large SVARs that can handle various cases where multiple linear equality and inequality restrictions are imposed on contemporaneous impulse responses, and can also address over-identifying restrictions.

This paper closely relates to the rapidly expanding literature on SVARs with equality and inequality restrictions (see, e.g., Baumeister and Hamilton, 2015, 2019; Arias et al., 2018; Amir-Ahmadi and Drautzburg, 2021; Bruns and Piffer, 2023). Well-known examples of these models are commonly known as Proxy-SVARs, which exploit information from external proxy variables in combination with additional inequality restrictions to identify structural shocks of interest. Therefore, this paper also contributes to the growing literature on Proxy-SVARs (see, e.g., Mertens and Ravn, 2013, 2014; Caldara and Herbst, 2019; Arias et al., 2021; Braun and Brüggemann, 2023). The paper by Arias et al. (2018) makes an important contribution by proposing an importance sampling approach for estimating Bayesian SVARs subject to equality and inequality restrictions. Their subsequent work (Arias et al., 2021) further extends their algorithm to accommodate the large number of zero restrictions inherent in the Proxy-SVAR framework. However, as they also noted, some practical concerns need careful attention when applying their importance sampling algorithm. First, it has been pointed out by Geweke (1989) that the validity of importance sampling requires the variance of the

☆ I would like to thank Joshua Chan, Jamie Cross, Liana Jacobi, Aubrey Poon, Yong Song, Kelly Trinh, Tomasz Woźniak and Dan Zhu, as well as Editor Serena Ng, an anonymous Associate Editor, and two anonymous reviewers for their constructive comments and valuable suggestions. I would also like to acknowledge financial support by the National Natural Science Foundation of China (72003064).

E-mail address: chenghan.hou@hotmail.com.

https://doi.org/10.1016/j.jeconom.2024.105850
Received 28 August 2022; Received in revised form 30 August 2024; Accepted 30 August 2024
Available online 20 September 2024
0304-4076/© 2024 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

importance weights to be finite. When the finite variance condition fails, the resulting importance sampling estimator might become unstable and strongly biased in practice. Second, their algorithm is computationally expensive when estimating high-dimensional models because it involves a numerical approximation step to evaluate the importance sampling weights (also known as volume elements). Third, the algorithm proposed by Arias et al. (2021) is based on the accept-reject algorithm of Rubio-Ramirez et al. (2010), which relies on random searches in parameter space through the use of draws of orthogonal matrices from the uniform (or Haar) distribution, and only those draws that lead to parameters satisfying all desired inequality restrictions are accepted for subsequent posterior analysis. Consequently, when the number of inequality restrictions is large, using the accept-reject algorithm to obtain a given number of posterior draws would quickly become computationally intractable. Caldara and Herbst (2019) and Braun and Brüggemann (2023) have proposed Metropolis–Hastings algorithms to estimate Proxy-SVARs. However, their algorithms might be computationally inefficient in high-dimensional settings.

To address these problems, we contribute to the literature by introducing an algorithm for estimating large Bayesian SVARs with linear equality and inequality restrictions. A key feature of our proposed algorithm is that it can handle SVARs with over-identifying restrictions, which is a case that the existing approaches, such as Arias et al. (2018, 2021) and Braun and Brüggemann (2023), cannot address. Our econometric framework is based on a B-model type SVAR (using the terminology of Lütkepohl (2005)) that has the reduced-form residuals $\mathbf{u}_t = \mathbf{B}\epsilon_t$, where ϵ_t is the vector of orthogonal structural shocks and $\mathbf{B}$ is the contemporaneous impact matrix of the structural shocks. We focus on cases in which a set of linear equality and inequality restrictions are imposed on the parameter space of $\mathbf{B}$. Our proposed algorithm is built upon a novel parameter transformation scheme, allowing us to efficiently draw each column of $\mathbf{B}$ conditional on the required linear equality and inequality restrictions.[1] The algorithm proposed in this paper can also be considered as an alternative approach to Waggoner and Zha (2003) for estimating SVARs. The difference is that their algorithm is designed for the A-model type SVARs (using the terminology of Lütkepohl (2005)), which do not incorporate any inequality restrictions. In contrast, our proposed algorithm is designed for B-model type SVARs and can handle a large number of linear inequality restrictions.

There are several benefits of working with a B-model type SVAR. First, information provided by economic theory about the reactions of endogenous variables to structural shocks is often incorporated into econometric analysis in the form of equality or inequality restrictions on the contemporaneous impact matrix $\mathbf{B}$. Second, under the framework of Proxy-SVARs, the equations for proxy variables in a B-model type SVAR have a more natural measurement error interpretation. Hence, directly characterizing prior belief on $\mathbf{B}$ is helpful for researchers to facilitate prior elicitation (Baumeister and Hamilton, 2015). Third, our econometric framework allows for the use of an independent prior on the reduced-form autoregressive coefficients and the contemporaneous impact matrix $\mathbf{B}$. This enables us to incorporate a variety of modern adaptive hierarchical shrinkage priors into our modeling framework that can effectively resolve the over-parameterization concerns in large models (Bhattacharya et al., 2015; Follett and Yu, 2017; Huber and Feldkircher, 2019; Cross et al., 2020; Chan, 2021).

This paper also relates to the recent works by Korobilis (2022) and Chan et al. (2022), who propose alternative approaches to the accept-reject algorithm of Rubio-Ramirez et al. (2010) for estimating SVARs with inequality restrictions. The main difference is that they assume the reduced-form residuals to have a factor error structure, while our modeling framework does not depend on this specific structure. Moreover, Korobilis (2022) and Chan et al. (2022) interpret the factors in their models as structural shocks and assume that the idiosyncratic shocks have negligible effects on their structural analysis. In contrast, the interpretation of structural shocks in our modeling framework is standard and consistent with the majority of SVAR studies.

We illustrate the usefulness of our methodology by using a large Bayesian Proxy-SVAR to revisit the empirical analysis in Stock and Watson (2012) that investigates the contributions of macroeconomic shocks to the fluctuations in GDP growth and employment during the 2007–09 recession. Our empirical study demonstrates the flexibility of our proposed method when multiple proxy variables are available for identifying multiple structural shocks. To be specific, we use 9 proxy variables constructed in the related studies to simultaneously identify 5 structural shocks — oil, monetary policy, productivity, liquidity and financial risk, and fiscal policy shocks — under a 30-variable Proxy-SVAR setting. We assume that each proxy variable is only correlated with the structural shock it is designed to identify, but uncorrelated to all the other remaining structural shocks. To the best of my knowledge, existing methods in the literature are not suitable for estimating such a high-dimensional Proxy-SVAR with this type of identifying assumption.[2] Our results show that liquidity and financial risk shocks play the most important role in driving the 2007–09 recession, contributing to about 50% of the decline in GDP growth and 30% of the decline in employment at 2009Q2. In contrast, the other macroeconomic shocks are relatively unimportant.

The remainder of the paper is organized as follows. Section 2 introduces our proposed algorithm. Section 3 provides an overview of Proxy-SVARs. Section 4 presents our empirical application, and Section 5 concludes.

[1] It is worth emphasizing that the algorithm proposed by Arias et al. (2018, 2021) can be applied to a broader class of priors than the method proposed in this paper. For example, their algorithm is applicable to a prior that assigns the same density to observationally equivalent structural parameters, a case where our proposed method cannot handle.

[2] In principle, the algorithm proposed by Caldara and Herbst (2019) can be applied to estimate Proxy-SVARs with this type of identifying assumption. However, their algorithm might become highly inefficient in practice when estimating high-dimensional models, such as the model used in the empirical application of this paper.

2. Methodology

This section develops a novel method for estimating SVARs conditional on linear equality and inequality restrictions. In Section 2.1, we describe our econometric framework, including model specification, prior distribution, and the linear equality and inequality restrictions of interest in this paper. Section 2.2 introduces a reparameterization and parameter transformation scheme. Sections 2.3 and 2.4 present our algorithm. Section 2.5 discusses some extensions and limitations of our proposed algorithm.

2.1. The framework

For expository purposes, we elaborate our methodology by working with a simplified version of SVAR with zero conditional mean:

$$\mathbf{y}_t = \mathbf{B}\boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n), \tag{1}$$

where $\mathbf{y}_t = (y_{1,t}, \dots, y_{n,t})'$ is an $n \times 1$ vector of endogenous variables, $\boldsymbol{\epsilon}_t$ is an $n \times 1$ vector of structural shocks and $\mathbf{B}$ is an $n \times n$ contemporaneous impact matrix that is assumed to be invertible. In this paper, we focus on cases where a set of linear restrictions are imposed on $\mathbf{B}$. Specifically, let $\mathbf{b}_i$ denote the ith column of $\mathbf{B}$ for $i = 1, \dots, n$. The linear equality and inequality restrictions of interest in this paper are given as follows:

$$\mathbf{R}_i^E \mathbf{b}_i = \mathbf{d}_i, \tag{2}$$

$$\mathbf{l}_i < \mathbf{R}_i^I \mathbf{b}_i < \mathbf{u}_i, \tag{3}$$

where $\mathbf{R}_i^E$ is $r_i^E \times n$, $\mathbf{d}_i$ is $r_i^E \times 1$, $\mathbf{R}_i^I$ is $r_i^I \times n$, and $\mathbf{l}_i$ and $\mathbf{u}_i$ are $r_i^I \times 1$. For the linear equality restrictions (2), without loss of generality, we assume $\text{rk}(\mathbf{R}_i^E) = r_i^E < n$. For the linear inequality restrictions (3), we assume $\mathbf{l}_i < \mathbf{u}_i$, and allow elements in $\mathbf{l}_i$ and $\mathbf{u}_i$ to be $-\infty$ or ∞ to indicate that some of the linear inequality restrictions are single-sided. All restrictions are assumed to be non-degenerate. Our framework for the linear restrictions specified in (2) and (3) is very flexible, accommodating various types of zero and sign restrictions commonly used in the SVAR literature. The only condition required is that the restricted $\mathbf{B}$ must be invertible.

We consider an independent Gaussian prior on $\mathbf{b}_i$, $i = 1, \dots, n$, with the support truncated to satisfy the linear restrictions (2) and (3):

$$\mathbf{b}_i \sim \mathcal{N}(\boldsymbol{\beta}_i, \mathbf{K}_i^{-1})\mathbf{1}(\mathbf{R}_i^E \mathbf{b}_i = \mathbf{d}_i)\mathbf{1}(\mathbf{l}_i < \mathbf{R}_i^I \mathbf{b}_i < \mathbf{u}_i), \tag{4}$$

where $\mathbf{1}(\cdot)$ is the indicator function. While our proposed methodology will be described under the independent truncated Gaussian prior, we will discuss in Section 2.5 that the technique developed in this paper can also be applied to several popular priors commonly used in the literature and can handle a more general form of linear restrictions.

The main objective of this paper is to develop an algorithm for drawing from the posterior distribution of $\mathbf{B}$. Since the posterior distribution of $\mathbf{B}$ is non-standard, direct sampling is not feasible. To obtain posterior draws of $\mathbf{B}$, this paper proposes an algorithm that draws each column of $\mathbf{B}$ conditional on the other columns. Specifically, our aim is to sequentially draw from $p(\mathbf{b}_i|\mathbf{B}_{-i}, \mathbf{y})$ where $\mathbf{B}_{-i} = (\mathbf{b}_1, \dots, \mathbf{b}_{i-1}, \mathbf{b}_{i+1}, \dots, \mathbf{b}_n)$, for $i = 1, \dots, n$. As we can permute any column of $\mathbf{B}$ to be updated to the first column, without loss of generality, we will present our algorithm for drawing from the conditional posterior $p(\mathbf{b}_1|\mathbf{B}_{-1}, \mathbf{y})$ in the following sections.

2.2. Reparameterization and parameter transformation

This section introduces a reparameterization scheme that transforms the SVAR given in (1) to facilitate drawing from the conditional posterior $p(\mathbf{b}_1|\mathbf{B}_{-1}, \mathbf{y})$. To set the stage, let $\mathbf{B}_{-1}^{\perp}$ be an $n \times 1$ vector that denotes the orthogonal complement of $\mathbf{B}_{-1}$ such that $\mathbf{B}_{-1}^{\perp\prime}\mathbf{B}_{-1} = \mathbf{0}_{1\times(n-1)}$. Our proposed method is based on reparameterizing the model using the following $n \times n$ matrix:

$$\mathbf{T} = \begin{pmatrix} \mathbf{B}_{-1}^{\perp\prime} \\ \mathbf{T}_{-1} \end{pmatrix}, \tag{5}$$

where $\mathbf{T}_{-1}$ is an $(n-1)\times n$ matrix such that $\mathbf{T}$ is invertible. When there are no restrictions imposed on the parameter space of $\mathbf{b}_1$, that is the case we will consider in Section 2.3, one can simply set $\mathbf{T}_{-1} = \mathbf{B}'_{-1}$. As we will discuss in Section 2.4, appropriately choosing $\mathbf{T}_{-1}$ can largely simplify the sampling problem when the linear equality and inequality restrictions are imposed on $\mathbf{b}_1$.

Premultiplying (1) by $\mathbf{T}$, we obtain the following reparameterized SVAR:

$$\widetilde{\mathbf{y}}_t = \widetilde{\mathbf{B}}\boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n), \tag{6}$$

where $\widetilde{\mathbf{y}}_t = \mathbf{T}\mathbf{y}_t$ and $\widetilde{\mathbf{B}} = \mathbf{T}\mathbf{B}$. Since both $\mathbf{T}$ and $\mathbf{B}$ are invertible, it follows that $\widetilde{\mathbf{B}}$ is also invertible. Let $\mathbf{w} = (w_1, \mathbf{w}_{-1})'$ denote the first column of $\widetilde{\mathbf{B}}$, where $\mathbf{w}_{-1} = (w_2, \dots, w_n)'$. The reparameterized SVAR given in (6) implicitly defines a parameter transformation that maps $\mathbf{b}_1$ to $\mathbf{w} = (w_1, \mathbf{w}'_{-1})'$:

$$\mathbf{w} = \mathbf{T}\mathbf{b}_1. \tag{7}$$

To draw from $p(\mathbf{b}_1|\mathbf{B}_{-1}, \mathbf{y})$, we propose a Gibbs sampler in the following sections that iteratively draws from the conditional posteriors $p(w_1|\mathbf{w}_{-1}, \mathbf{B}_{-1}, \widetilde{\mathbf{y}})$ and $p(\mathbf{w}_{-1}|w_1, \mathbf{B}_{-1}, \widetilde{\mathbf{y}})$. Since (7) defines a one-to-one mapping between $\mathbf{w}$ and $\mathbf{b}_1$, posterior draws of $\mathbf{b}_1$ can be obtained by transforming the draws of $\mathbf{w} = (w_1, \mathbf{w}_{-1})$ via the inverse mapping $\mathbf{b}_1 = \mathbf{T}^{-1}\mathbf{w}$.

2.3. Without linear equality and inequality restrictions

As discussed in the previous section, draws from $p(\mathbf{b}_1|\mathbf{B}_{-1},\mathbf{y})$ can be obtained by iteratively drawing from the conditional posteriors $p(w_1|\mathbf{w}_{-1},\mathbf{B}_{-1},\widetilde{\mathbf{y}})$ and $p(\mathbf{w}_{-1}|w_1,\mathbf{B}_{-1},\widetilde{\mathbf{y}})$. In this section, we provide the results for these conditional posteriors in the case when there are no restrictions imposed on $\mathbf{b}_1$. Although this is not the main interest of this paper, some results presented in this section are instructive for later reference when we consider the linear equality and inequality restrictions (2) and (3) in Section 2.4.

By construction, the transformation $\mathbf{T}$ described in (5) does not depend on $\mathbf{b}_1$. Thus, the joint likelihood function of $\mathbf{w}=(w_1,\mathbf{w}_{-1})$ conditional on $\mathbf{B}_{-1}$ is given by

$$p(\widetilde{\mathbf{y}}|w_1,\mathbf{w}_{-1},\mathbf{B}_{-1}) \propto |\det(\widetilde{\mathbf{B}})|^{-T} e^{-\frac{1}{2}\mathrm{vec}(\widetilde{\mathbf{B}}^{-1})'\mathbf{D}\mathrm{vec}(\widetilde{\mathbf{B}}^{-1})}, \tag{8}$$

where $\widetilde{\mathbf{y}}=(\widetilde{\mathbf{y}}_1',\ldots,\widetilde{\mathbf{y}}_T')'$, $|\det(\widetilde{\mathbf{B}})|$ denotes the absolute value of the determinant of $\widetilde{\mathbf{B}}$, and $\mathbf{D}=\widetilde{\mathbf{Y}}'\widetilde{\mathbf{Y}}\otimes\mathbf{I}_n$ with $\widetilde{\mathbf{Y}}=(\widetilde{\mathbf{y}}_1,\ldots,\widetilde{\mathbf{y}}_T)'$. Given this joint likelihood function for $\mathbf{w}=(w_1,\mathbf{w}_{-1}')'$, the following lemma provides the results for the conditional likelihoods with respect to w_1 and $\mathbf{w}_{-1}$.

Lemma 1. *Given the joint likelihood function (8) for $\mathbf{w}=(w_1,\mathbf{w}_{-1}')'$, the conditional likelihood of $\mathbf{w}_{-1}$ is given by*

$$p_{\mathbf{w}_{-1}}(\widetilde{\mathbf{y}}|w_1,\mathbf{w}_{-1},\mathbf{B}_{-1}) \propto e^{-\frac{1}{2}(\mathbf{q}_{\mathbf{w}_{-1}}-\mathbf{Z}_{\mathbf{w}_{-1}}\mathbf{w}_{-1})'(\mathbf{q}_{\mathbf{w}_{-1}}-\mathbf{Z}_{\mathbf{w}_{-1}}\mathbf{w}_{-1})}, \tag{9}$$

and the conditional likelihood of w_1 is given by

$$p_{w_1}(\widetilde{\mathbf{y}}|w_1,\mathbf{w}_{-1},\mathbf{B}_{-1}) \propto |w_1|^{-T} e^{-\frac{1}{2}(w_1^{-1}\gamma_1+w_1^{-2}\gamma_2)}. \tag{10}$$

The proof and the expressions of $\mathbf{q}_{\mathbf{w}_{-1}}$, $\mathbf{Z}_{\mathbf{w}_{-1}}$, γ_1, and γ_2 are provided in Appendix A.2.

Since $\mathbf{w}=\mathbf{T}\mathbf{b}_1$ and $\mathbf{b}_1$ follows a Gaussian distribution, when $\mathbf{b}_1$ is not subject to any restrictions, a standard property of the Gaussian distribution shows that the conditional priors for w_1 and $\mathbf{w}_{-1}$ are also Gaussian distributions:

$$(\mathbf{w}_{-1}|w_1) \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{w}_{-1}},\boldsymbol{\Lambda}_{\mathbf{w}_{-1}}^{-1}), \tag{11}$$

$$(w_1|\mathbf{w}_{-1}) \sim \mathcal{N}(\mu_{w_1},\lambda_{w_1}^{-1}), \tag{12}$$

where the expressions of $\boldsymbol{\mu}_{\mathbf{w}_{-1}}$, $\boldsymbol{\Lambda}_{\mathbf{w}}$, μ_{w_1} and λ_{w_1} are given in Appendix A.1. By combining these conditional priors with the conditional likelihoods provided in Lemma 1, we obtain the following conditional posteriors for $\mathbf{w}_{-1}$ and w_1:

$$\begin{aligned}
p(\mathbf{w}_{-1}|w_1,\mathbf{B}_{-1},\widetilde{\mathbf{y}}) &\propto p_{\mathbf{w}_{-1}}(\widetilde{\mathbf{y}}|w_1,\mathbf{w}_{-1},\mathbf{B}_{-1})p(\mathbf{w}_{-1}|w_1)\\
&\propto e^{-\frac{1}{2}(\mathbf{q}_{\mathbf{w}_{-1}}-\mathbf{Z}_{\mathbf{w}_{-1}}\mathbf{w}_{-1})'(\mathbf{q}_{\mathbf{w}_{-1}}-\mathbf{Z}_{\mathbf{w}_{-1}}\mathbf{w}_{-1})}e^{-\frac{1}{2}(\mathbf{w}_{-1}-\boldsymbol{\mu}_{\mathbf{w}_{-1}})'\boldsymbol{\Lambda}_{\mathbf{w}_{-1}}(\mathbf{w}_{-1}-\boldsymbol{\mu}_{\mathbf{w}_{-1}})}\\
&\propto e^{-\frac{1}{2}(\mathbf{w}_{-1}'\widehat{\mathbf{K}}_{\mathbf{w}_{-1}}\mathbf{w}_{-1}-2\mathbf{w}_{-1}'\widehat{\mathbf{K}}_{\mathbf{w}_{-1}}\widehat{\mathbf{w}}_{-1})},\\
p(w_1|\mathbf{w}_{-1},\mathbf{B}_{-1},\widetilde{\mathbf{y}}) &\propto p_{w_1}(\widetilde{\mathbf{y}}|w_1,\mathbf{w}_{-1},\mathbf{B}_{-1})p(w_1|\mathbf{w}_{-1})\\
&\propto |w_1|^{-T}e^{-\frac{1}{2}(w_1^{-1}\gamma_1+w_1^{-2}\gamma_2)}e^{-\frac{\lambda_{w_1}}{2}(w_1-\mu_{w_1})^2}.
\end{aligned}$$

where $\widehat{\mathbf{K}}_{\mathbf{w}_{-1}}=\mathbf{Z}_{\mathbf{w}_{-1}}'\mathbf{Z}_{\mathbf{w}_{-1}}+\boldsymbol{\Lambda}_{\mathbf{w}_{-1}}$ and $\widehat{\mathbf{w}}_{-1}=\widehat{\mathbf{K}}_{\mathbf{w}_{-1}}^{-1}\left(\mathbf{Z}_{\mathbf{w}_{-1}}'\mathbf{q}_{\mathbf{w}_{-1}}+\boldsymbol{\Lambda}_{\mathbf{w}_{-1}}\boldsymbol{\mu}_{\mathbf{w}_{-1}}\right)$.

The conditional posteriors of $\mathbf{w}_{-1}$ and w_1 presented above serve as the key building blocks of our proposed algorithm, and we summarize these results in Proposition 1.

Proposition 1. *Given the reparameterized SVAR (6) with the priors (11) and (12), the density of the conditional posterior of $\mathbf{w}_{-1}$ is given by*

$$p(\mathbf{w}_{-1}|w_1,\mathbf{B}_{-1},\widetilde{\mathbf{y}}) \propto e^{-\frac{1}{2}(\mathbf{w}_{-1}'\widehat{\mathbf{K}}_{\mathbf{w}_{-1}}\mathbf{w}_{-1}-2\mathbf{w}_{-1}'\widehat{\mathbf{K}}_{\mathbf{w}_{-1}}\widehat{\mathbf{w}}_{-1})},$$

and the density of the conditional posterior of w_1 is given by

$$p(w_1|\mathbf{w}_{-1},\mathbf{B}_{-1},\widetilde{\mathbf{y}}) \propto |w_1|^{-T}e^{-\frac{1}{2}(w_1^{-1}\gamma_1+w_1^{-2}\gamma_2)}e^{-\frac{\lambda_{w_1}}{2}(w_1-\mu_{w_1})^2}.$$

Proposition 1 shows that the conditional posterior of $\mathbf{w}_{-1}$ follows a Gaussian distribution

$$(\mathbf{w}_{-1}|w_1,\mathbf{B}_{-1},\widetilde{\mathbf{y}}) \sim \mathcal{N}(\widehat{\mathbf{w}}_{-1},\widehat{\mathbf{K}}_{\mathbf{w}_{-1}}^{-1}).$$

However, the conditional posterior $p(w_1|\mathbf{w}_{-1},\mathbf{B}_{-1},\widetilde{\mathbf{y}})$ is not a standard distribution. Therefore, we propose an independent Metropolis–Hastings step for drawing from $p(w_1|\mathbf{w}_{-1},\mathbf{B}_{-1},\widetilde{\mathbf{y}})$. Specifically, we suggest to use a distribution that has the density function

$$q^*(w_1^*) \propto |w_1^*|^{-T}e^{-\frac{1}{2}(w_1^{*-1}\gamma_1+w_1^{*-2}\gamma_2)} \tag{13}$$

as our proposal distribution. Proposition 2 provides an efficient method for drawing from this proposal distribution.

Proposition 2. *Suppose* $\widetilde{w}_1 \sim \mathcal{AN}\left(-\frac{\gamma_1}{2\sqrt{\gamma_2(T-2)}}, \frac{1}{T-2}\right)$, *where* $\mathcal{AN}(a,b)$ *denotes the absolute normal distribution that has the density function*

$$f_{AN}(x;a,b) \propto |x|^{\frac{1}{b}} e^{-\frac{1}{2b}(x-a)^2}, \quad x \in \mathbb{R}, b \in \mathbb{R}^+, a \in \mathbb{R}.$$

Then $w_1^* = \widetilde{w}_1^{-1}\sqrt{\frac{\gamma_2}{T-2}}$ *follows a distribution with density function (13). The proof is provided in Appendix A.3.*

To apply Proposition 2, we need a method to draw from the absolute normal distribution $\mathcal{AN}\left(-\frac{\gamma_1}{2\sqrt{\gamma_2(T-2)}}, \frac{1}{T-2}\right)$. This can be done by using the normal mixture approximation proposed in Appendix C of Villani (2009) (see also Section 2.5.4 for more discussion). Thus, given a current draw of $\mathbf{w} = (w_1, \mathbf{w}_{-1}')'$, a proposal draw $w_1^* \sim q^*(w_1^*)$ is accepted with probability $\min\left\{1, \frac{e^{-\frac{\lambda_{w_1}}{2}(w_1^*-\mu_{w_1})^2}}{e^{-\frac{\lambda_{w_1}}{2}(w_1-\mu_{w_1})^2}}\right\}$. We summarize our algorithm for drawing $\mathbf{b}_1$ without restrictions in Algorithm 1.

Algorithm 1 Drawing from $p(\mathbf{b}_1|\mathbf{B}_{-1},\mathbf{y})$ without restrictions.

Given $\mathbf{b}_1$, complete the following steps:

1. Set $\mathbf{w} = (w_1, \mathbf{w}_{-1}')' = \mathbf{T}\mathbf{b}_1$ and $\widetilde{\mathbf{y}} = (\widetilde{\mathbf{y}}_1', \dots, \widetilde{\mathbf{y}}_T')'$, where $\mathbf{T}$ is given in (5) and $\widetilde{\mathbf{y}}_t = \mathbf{T}\mathbf{y}_t$.
2. Draw $(\mathbf{w}_{-1}|w_1, \mathbf{B}_{-1}, \widetilde{\mathbf{y}}) \sim \mathcal{N}(\widehat{\mathbf{w}}_{-1}, \widehat{\mathbf{K}}_{\mathbf{w}_{-1}}^{-1})$.
3. Draw a proposal $w_1^* \sim q^*(w_1^*) \propto |w_1^*|^{-T} e^{-\frac{1}{2}(w_1^{*-1}\gamma_1 + w_1^{*-2}\gamma_2)}$ using the method proposed in Proposition 2. Set $w_1 = w_1^*$ with probability $\min\left\{1, \frac{e^{-\frac{\lambda_{w_1}}{2}(w_1^*-\mu_{w_1})^2}}{e^{-\frac{\lambda_{w_1}}{2}(w_1-\mu_{w_1})^2}}\right\}$.
4. Set $\mathbf{w} = (w_1, \mathbf{w}_{-1}')'$ and return $\mathbf{b}_1 = \mathbf{T}^{-1}\mathbf{w}$.

2.4. With linear equality and inequality restrictions

This section extends the method outlined in Section 2.3 to the case when the linear restrictions are imposed. In particular, we now consider drawing from $p(\mathbf{b}_1|\mathbf{B}_{-1},\mathbf{y})$ with the linear restrictions (2) and (3) for $\mathbf{b}_1$, i.e., $\mathbf{R}_1^E\mathbf{b}_1 = \mathbf{d}_1$ and $\mathbf{l}_1 < \mathbf{R}_1^I\mathbf{b}_1 < \mathbf{u}_1$.

We first discuss how one can appropriately choose the submatrix $\mathbf{T}_{-1}$ in the transformation (5) to simplify the procedure of drawing $\mathbf{b}_1$. Let $\mathbf{T}_{-1} = (\mathbf{T}_{-1}^{*\prime}, \mathbf{R}_1^{E\prime})'$, where $\mathbf{T}_{-1}^*$ can be any $(n - r_1^E - 1)\times n$ matrix such that the resulting transformation $\mathbf{T}$ is invertible, then the transformation (5) now can be written as:

$$\mathbf{T} = \begin{pmatrix} \mathbf{B}_{-1}^{\perp\prime} \\ \mathbf{T}_{-1}^* \\ \mathbf{R}_1^E \end{pmatrix}. \tag{14}$$

Note that the invertibility of $\mathbf{T}$ also requires that the matrix $(\mathbf{B}_{-1}^{\perp}, \mathbf{R}_1^{E\prime})'$ is of full row rank, i.e., $\mathrm{rk}[(\mathbf{B}_{-1}^{\perp}, \mathbf{R}_1^{E\prime})'] = r_1^E + 1$. Throughout this section, we will describe our algorithm under this assumption, while in Appendix A.4 we demonstrate that this assumption can be relaxed by appropriately reformulating the linear restrictions $\mathbf{R}_1^E\mathbf{b}_1 = \mathbf{d}_1$.

If we partition $\mathbf{w}_{-1} = (\mathbf{w}_{-1,1}', \mathbf{w}_{-1,2}')'$ where $\mathbf{w}_{-1,1} = (w_2, \dots, w_{n-r_1^E})'$ and $\mathbf{w}_{-1,2} = (w_{n-r_1^E+1}, \dots, w_n)'$, with the transformation specified in (14), we can rewrite (7) as

$$\mathbf{T}\mathbf{b}_1 = \begin{pmatrix} \mathbf{B}_{-1}^{\perp\prime} \\ \mathbf{T}_{-1}^* \\ \mathbf{R}_1^E \end{pmatrix}\mathbf{b}_1 = \mathbf{w} = \begin{pmatrix} w_1 \\ \mathbf{w}_{-1} \end{pmatrix} = \begin{pmatrix} w_1 \\ \mathbf{w}_{-1,1} \\ \mathbf{w}_{-1,2} \end{pmatrix} = \begin{pmatrix} w_1 \\ \mathbf{w}_{-1,1} \\ \mathbf{d}_1 \end{pmatrix}. \tag{15}$$

The last equality holds because $\mathbf{R}_1^E\mathbf{b}_1 = \mathbf{d}_1$. The key benefit of working with the transformed parameters $\mathbf{w}$, as obtained in (15), is that it simplifies the problem of drawing $\mathbf{b}_1$ with the linear restrictions $\mathbf{R}_1^E\mathbf{b}_1 = \mathbf{d}_1$ to a problem of drawing $\mathbf{w} = (w_1, \mathbf{w}_{-1,1}', \mathbf{w}_{-1,2}')'$ with $\mathbf{w}_{-1,2} = \mathbf{d}_1$. Hence, with the parameter transformation (15), the only free parameters in $\mathbf{w}$ that need to be sampled are $(w_1, \mathbf{w}_{-1,1}')'$.

Given the truncated Gaussian prior (4) for $\mathbf{b}_1$, we show in Appendix A.1 that the conditional priors for $\mathbf{w}_{-1,1}$ and w_1 are also truncated Gaussian distributions:

$$(\mathbf{w}_{-1,1}|w_1, \mathbf{w}_{-1,2}) \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{w}_{-1,1}}, \boldsymbol{\Lambda}_{\mathbf{w}_{-1,1}}^{-1})\mathbf{1}(\mathbf{l}_{\mathbf{w}_{-1,1}} < \mathbf{R}_{\mathbf{w}_{-1,1}}^I\mathbf{w}_{-1,1} < \mathbf{u}_{\mathbf{w}_{-1,1}}), \tag{16}$$

$$(w_1|\mathbf{w}_{-1}) \sim \mathcal{N}(\mu_{w_1}, \lambda_{w_1}^{-1})\mathbf{1}(\mathbf{l}_{w_1} < \mathbf{R}_{w_1}^I w_1 < \mathbf{u}_{w_1}), \tag{17}$$

where $\mathbf{R}_{\mathbf{w}_{-1,1}}^I$ is $r_1^I \times (n - r_1^E - 1)$, and $\mathbf{l}_{\mathbf{w}_{-1,1}}$, $\mathbf{u}_{\mathbf{w}_{-1,1}}$, $\mathbf{R}_{w_1}^I$, $\mathbf{l}_{w_1}$ and $\mathbf{u}_{w_1}$ are $r_1^I \times 1$. Moreover, $(\mathbf{R}_{\mathbf{w}_{-1,1}}^I, \mathbf{l}_{\mathbf{w}_{-1,1}}, \mathbf{u}_{\mathbf{w}_{-1,1}})$ does not depend on $\mathbf{w}_{-1,1}$ and $(\mathbf{R}_{w_1}^I, \mathbf{l}_{w_1}, \mathbf{u}_{w_1})$ does not depend on w_1. These conditional priors can then be used in combination with the conditional likelihoods (9) and (10) given in Lemma 1 to obtain the conditional posteriors of $\mathbf{w}_{-1,1}$ and w_1.

To get the conditional posterior of $\mathbf{w}_{-1,1}$, we first rewrite $\mathbf{w}_{-1} = (\mathbf{w}'_{-1,1}, \mathbf{d}'_1)'$ as $\mathbf{w}_{-1} = \mathbf{S}_1\mathbf{w}_{-1,1} + \mathbf{S}_2\mathbf{d}_1$, where $\mathbf{S}_1$ and $\mathbf{S}_2$ are selection matrices with dimensions $(n-1)\times(n-r_1^E-1)$ and $(n-1)\times r_1^E$, respectively. In particular, we have that $(\mathbf{S}_1, \mathbf{S}_2)$ is an $(n-1)$-dimensional identity matrix. Substituting $\mathbf{w}_{-1} = \mathbf{S}_1\mathbf{w}_{-1,1} + \mathbf{S}_2\mathbf{d}_1$ into (9), we obtain the conditional likelihood of $\mathbf{w}_{-1,1}$:

$$\begin{aligned} p_{\mathbf{w}_{-1,1}}(\widetilde{\mathbf{y}}|w_1, \mathbf{w}_{-1,1}, \mathbf{B}_{-1}) &\propto e^{-\frac{1}{2}(\mathbf{q}_{\mathbf{w}_{-1}} - \mathbf{Z}_{\mathbf{w}_{-1}}(\mathbf{S}_1\mathbf{w}_{-1,1} + \mathbf{S}_2\mathbf{d}_1))'(\mathbf{q}_{\mathbf{w}_{-1}} - \mathbf{Z}_{\mathbf{w}_{-1}}(\mathbf{S}_1\mathbf{w}_{-1,1} + \mathbf{S}_2\mathbf{d}_1))} \\ &\propto e^{-\frac{1}{2}(\mathbf{q}_{\mathbf{w}_{-1,1}} - \mathbf{Z}_{\mathbf{w}_{-1,1}}\mathbf{w}_{-1,1})'(\mathbf{q}_{\mathbf{w}_{-1,1}} - \mathbf{Z}_{\mathbf{w}_{-1,1}}\mathbf{w}_{-1,1})}, \end{aligned} \tag{18}$$

where $\mathbf{q}_{\mathbf{w}_{-1,1}} = \mathbf{q}_{\mathbf{w}_{-1}} - \mathbf{Z}_{\mathbf{w}_{-1}}\mathbf{S}_2\mathbf{d}_1$ and $\mathbf{Z}_{\mathbf{w}_{-1,1}} = \mathbf{Z}_{\mathbf{w}_{-1}}\mathbf{S}_1$. Combining this conditional likelihood with the conditional prior (16) gives the conditional posterior of $\mathbf{w}_{-1,1}$:

$$\begin{aligned} p(\mathbf{w}_{-1,1}|w_1, \mathbf{w}_{-1,2}, \mathbf{B}_{-1}, \widetilde{\mathbf{y}}) &\propto p_{\mathbf{w}_{-1,1}}(\widetilde{\mathbf{y}}|w_1, \mathbf{w}_{-1,1}, \mathbf{B}_{-1})p(\mathbf{w}_{-1,1}|w_1, \mathbf{w}_{-1,2}) \\ &\propto e^{-\frac{1}{2}(\mathbf{w}'_{-1,1}\widehat{\mathbf{K}}_{\mathbf{w}_{-1,1}}\mathbf{w}_{-1,1} - 2\mathbf{w}'_{-1,1}\widehat{\mathbf{K}}_{\mathbf{w}_{-1,1}}\widehat{\mathbf{w}}_{-1,1})}\mathbf{1}(\mathbf{l}_{\mathbf{w}_{-1,1}} < \mathbf{R}^I_{\mathbf{w}_{-1,1}}\mathbf{w}_{-1,1} < \mathbf{u}_{\mathbf{w}_{-1,1}}), \end{aligned}$$

where $\widehat{\mathbf{K}}_{\mathbf{w}_{-1,1}} = \mathbf{Z}'_{\mathbf{w}_{-1,1}}\mathbf{Z}_{\mathbf{w}_{-1,1}} + \boldsymbol{\Lambda}_{\mathbf{w}_{-1,1}}$ and $\widehat{\mathbf{w}}_{-1,1} = \widehat{\mathbf{K}}^{-1}_{\mathbf{w}_{-1,1}}\left(\mathbf{Z}'_{\mathbf{w}_{-1,1}}\mathbf{q}_{\mathbf{w}_{-1,1}} + \boldsymbol{\Lambda}_{\mathbf{w}_{-1,1}}\boldsymbol{\mu}_{\mathbf{w}_{-1,1}}\right)$. This implies that the conditional posterior of $\mathbf{w}_{-1,1}$ is a Gaussian distribution with linear inequality restrictions $\mathbf{l}_{\mathbf{w}_{-1,1}} < \mathbf{R}^I_{\mathbf{w}_{-1,1}}\mathbf{w}_{-1,1} < \mathbf{u}_{\mathbf{w}_{-1,1}}$:

$$(\mathbf{w}_{-1,1}|w_1, \mathbf{w}_{-1,2}, \mathbf{B}_{-1}, \widetilde{\mathbf{y}}) \sim \mathcal{N}(\widehat{\mathbf{w}}_{-1,1}, \widehat{\mathbf{K}}^{-1}_{\mathbf{w}_{-1,1}})\mathbf{1}(\mathbf{l}_{\mathbf{w}_{-1,1}} < \mathbf{R}^I_{\mathbf{w}_{-1,1}}\mathbf{w}_{-1,1} < \mathbf{u}_{\mathbf{w}_{-1,1}}).$$

To obtain the conditional posterior of w_1, we first note that the likelihood of w_1 conditional on $\mathbf{w}_{-1} = (\mathbf{w}'_{-1,1}, \mathbf{d}'_1)'$ is exactly the same as (10). Thus, with the conditional prior (17), the conditional posterior of w_1 is given by

$$p(w_1|\mathbf{w}_{-1}, \mathbf{B}_{-1}, \widetilde{\mathbf{y}}) \propto |w_1|^{-T}e^{-\frac{1}{2}(w_1^{-1}\gamma_1 + w_1^{-2}\gamma_2)}e^{-\frac{\lambda_{w_1}}{2}(w_1-\mu_{w_1})^2}\mathbf{1}(\mathbf{l}_{w_1} < \mathbf{R}^I_{w_1}w_1^* < \mathbf{u}_{w_1}).$$

The conditional posterior $p(w_1|\mathbf{w}_{-1}, \mathbf{B}_{-1}, \widetilde{\mathbf{y}})$ presented above is the same as the one given in Proposition 1, except that the parameter space of w_1 is now restricted by the inequality restrictions $\mathbf{l}_{w_1} < \mathbf{R}^I_{w_1}w_1 < \mathbf{u}_{w_1}$. Therefore, we can draw from $p(w_1|\mathbf{w}_{-1}, \mathbf{B}_{-1}, \widetilde{\mathbf{y}})$ via an independent Metropolis–Hastings step by first drawing a proposal w_1^* using the method proposed in Proposition 2, and then set $w_1 = w_1^*$ with probability $\min\left\{1, \frac{e^{-\frac{\lambda_{w_1}}{2}(w_1^*-\mu_{w_1})^2}}{e^{-\frac{\lambda_{w_1}}{2}(w_1-\mu_{w_1})^2}}\right\}\mathbf{1}(\mathbf{l}_{w_1} < \mathbf{R}^I_{w_1}w_1^* < \mathbf{u}_{w_1})$. We summarize our algorithm for drawing $\mathbf{b}_1$ conditional on the linear restrictions $\mathbf{R}^E_1\mathbf{b}_1 = \mathbf{d}_1$ and $\mathbf{l}_1 < \mathbf{R}^I_1\mathbf{b}_1 < \mathbf{u}_1$ in Algorithm 2.

Algorithm 2 Drawing from $p(\mathbf{b}_1|\mathbf{B}_{-1}, \mathbf{y})$ with $\mathbf{R}^E_1\mathbf{b}_1 = \mathbf{d}_1$ and $\mathbf{l}_1 < \mathbf{R}^I_1\mathbf{b}_1 < \mathbf{u}_1$.

Given $\mathbf{b}_1$, complete the following steps:

1. Set $\mathbf{w} = (w_1, \mathbf{w}'_{-1,1}, \mathbf{w}'_{-1,2})' = \mathbf{T}\mathbf{b}_1$ and $\widetilde{\mathbf{y}} = (\widetilde{\mathbf{y}}'_1, \ldots, \widetilde{\mathbf{y}}'_T)'$, where $\mathbf{T}$ is given in (14), $\widetilde{\mathbf{y}}_t = \mathbf{T}\mathbf{y}_t$ and $\mathbf{w}_{-1,2} = \mathbf{d}_1$.
2. Draw $(\mathbf{w}_{-1,1}|w_1, \mathbf{w}_{-1,2}, \mathbf{B}_{-1}, \widetilde{\mathbf{y}}) \sim \mathcal{N}(\widehat{\mathbf{w}}_{-1,1}, \widehat{\mathbf{K}}^{-1}_{\mathbf{w}_{-1,1}})\mathbf{1}(\mathbf{l}_{\mathbf{w}_{-1,1}} < \mathbf{R}^I_{\mathbf{w}_{-1,1}}\mathbf{w}_{-1,1} < \mathbf{u}_{\mathbf{w}_{-1,1}})$.
3. Draw a proposal $w_1^* \sim q^*(w_1^*) \propto |w_1^*|^{-T}e^{-\frac{1}{2}(w_1^{*-1}\gamma_1 + w_1^{*-2}\gamma_2)}$ using the method proposed in Proposition 2. Set $w_1 = w_1^*$ with probability $\min\left\{1, \frac{e^{-\frac{\lambda_{w_1}}{2}(w_1^*-\mu_{w_1})^2}}{e^{-\frac{\lambda_{w_1}}{2}(w_1-\mu_{w_1})^2}}\right\}\mathbf{1}(\mathbf{l}_{w_1} < \mathbf{R}^I_{w_1}w_1^* < \mathbf{u}_{w_1})$.
4. Set $\mathbf{w} = (w_1, \mathbf{w}'_{-1}, \mathbf{w}'_{-1,2})'$ and return $\mathbf{b}_1 = \mathbf{T}^{-1}\mathbf{w}$.

Step 2 of Algorithm 2 draws $\mathbf{w}_{-1,1}$ from a Gaussian distribution with linear inequality restrictions. This can be done by using the algorithm of Botev (2017) when the number of restrictions (number of rows of $\mathbf{R}^I_{\mathbf{w}_{-1,1}}$) is less than or equal to the dimension of $\mathbf{w}_{-1,1}$, i.e., $r_1^I \le n - r_1^E - 1$. Otherwise, when the number of restrictions exceeds the dimension of $\mathbf{w}_{-1,1}$, i.e., $r_1^I > n - r_1^E - 1$, draws of $\mathbf{w}_{-1,1}$ can be obtained using the Gibbs sampler proposed by Rodriguez-Yam et al. (2004).[3] Note that Algorithm 2 makes draws of $\mathbf{b}_1$ conditional on the linear inequality restrictions without relying on the conventional accept-reject algorithm of Rubio-Ramirez et al. (2010). This could improve computational efficiency, especially when dealing with a large number of inequality restrictions.

2.5. Extensions and limitations

This section discusses some extensions and limitations of our proposed algorithm.

2.5.1. General linear restrictions on B

Our proposed algorithm presented in Section 2.4 considers the case when linear restrictions are independently imposed on each column of $\mathbf{B}$. In this section, we demonstrate that our methodology can be extended to handle a more general form of linear restrictions:

$$\bar{\mathbf{R}}^E\mathrm{vec}(\mathbf{B}) = \bar{\mathbf{d}}, \tag{19}$$

[3] While the Gibbs sampler of Rodriguez-Yam et al. (2004) is also applicable when $r_1^I \le n - r_1^E - 1$, we prefer the direct sampling approach proposed by Botev (2017) due to its better mixing properties.

$$\widetilde{\mathbf{l}} < \widetilde{\mathbf{R}}^I \mathrm{vec}(\mathbf{B}) < \widetilde{\mathbf{u}}, \tag{20}$$

where $\bar{\mathbf{R}}^I$ is $\bar{r}^E \times n^2$, $\bar{\mathbf{d}}$ is $\bar{r}^E \times 1$, $\widetilde{\mathbf{R}}^I$ is $\widetilde{r}^I \times n^2$, and $\widetilde{\mathbf{l}}$ and $\widetilde{\mathbf{u}}$ are $\widetilde{r}^I \times 1$. It can be seen that the linear restrictions specified in (19) and (20) nest the linear restrictions (2) and (3) as a special case, which also include many identifying restrictions commonly used in SVAR studies, such as ranking restrictions on the contemporaneous impulse responses (Amir-Ahmadi and Drautzburg, 2021).

The key is to realize that (19) and (20) can be reformulated into a form similar to (2) and (3). To see this, we first write $\bar{\mathbf{R}}^E = (\bar{\mathbf{R}}_1^E, \dots, \bar{\mathbf{R}}_n^E)$ and $\widetilde{\mathbf{R}}^I = (\widetilde{\mathbf{R}}_1^I, \dots, \widetilde{\mathbf{R}}_n^I)$, where $\bar{\mathbf{R}}_i^E$ is $\bar{r}^E \times n$ and $\widetilde{\mathbf{R}}_i^I$ is $\widetilde{r}^I \times n$. We then define $\bar{\mathbf{R}}_{-i}^E = (\bar{\mathbf{R}}_1^E, \dots, \bar{\mathbf{R}}_{i-1}^E, \bar{\mathbf{R}}_{i+1}^E, \dots, \bar{\mathbf{R}}_n^E)$ and $\widetilde{\mathbf{R}}_{-i}^I = (\widetilde{\mathbf{R}}_1^I, \dots, \widetilde{\mathbf{R}}_{i-1}^I, \widetilde{\mathbf{R}}_{i+1}^I, \dots, \widetilde{\mathbf{R}}_n^I)$, for $i = 1, \dots, n$. Thus, (19) and (20) can be rewritten as

$$\bar{\mathbf{R}}_i^E \mathbf{b}_i = \bar{\mathbf{d}}_i, \tag{21}$$

$$\widetilde{\mathbf{l}}_i < \widetilde{\mathbf{R}}_i^I \mathbf{b}_i < \widetilde{\mathbf{u}}_i, \tag{22}$$

for $i = 1, \dots, n$, where $\bar{\mathbf{d}} = \bar{\mathbf{d}}_i - \bar{\mathbf{R}}_{-i}^E \mathrm{vec}(\mathbf{B}_{-i})$, $\widetilde{\mathbf{l}}_i = \widetilde{\mathbf{l}} - \widetilde{\mathbf{R}}_{-i}^I \mathrm{vec}(\mathbf{B}_{-i})$ and $\widetilde{\mathbf{u}}_i = \widetilde{\mathbf{u}} - \widetilde{\mathbf{R}}_{-i}^I \mathrm{vec}(\mathbf{B}_{-i})$. It can be seen that the linear inequality restrictions (22) are of the same form as (3). For the linear equality restrictions (21), we need to further assume that $\mathrm{rk}(\bar{\mathbf{R}}_i^E) = \widetilde{r}_i^E < n$. In other words, we only allow $\widetilde{r}_i^E < n$ independent linear equality restrictions. Consequently, there exists a $\widetilde{r}_i^E \times n$ matrix $\widetilde{\mathbf{R}}_i^E$ with $\mathrm{rk}(\widetilde{\mathbf{R}}_i^E) = \widetilde{r}_i^E < n$, such that (21) can be equivalently expressed as

$$\widetilde{\mathbf{R}}_i^E \mathbf{b}_i = \widetilde{\mathbf{d}}_i, \tag{23}$$

where $\widetilde{\mathbf{d}}_i$ is $\widetilde{r}_i^E \times 1$, $i = 1, \dots, n$.[4] We can see that the reformulated equality restrictions (23) have the same form as those given in (2). Note that, in (22) and (23), $\widetilde{\mathbf{R}}_i^E$, $\widetilde{\mathbf{d}}_i$, $\widetilde{\mathbf{R}}_i^I$, $\widetilde{\mathbf{l}}_i$ and $\widetilde{\mathbf{u}}_i$ depend only on $\mathbf{B}_{-i}$, but not on $\mathbf{b}_i$. Hence, the methodology developed in Section 2.4 can be applied straightforwardly to draw $\mathbf{b}_i$ conditional on $\mathbf{B}_{-i}$ by replacing (2) and (3) with (22) and (23), respectively.

2.5.2. Additional nonlinear inequality restrictions

Our proposed algorithm can also be extended to incorporate additional inequality restrictions on nonlinear functions of the model parameters. Examples of such functions include impulse responses at higher horizons, historical decompositions, forecast error variance decompositions, and the reliability condition of proxy variables in Proxy-SVAR analysis. This extension can be achieved by simply adding accept-reject steps to our proposed sampler.[5] In many applications of SVARs or Proxy-SVARs with inequality restrictions, the number of nonlinear inequality restrictions is often smaller than the number of linear inequality restrictions on $\mathbf{B}$. Since our proposed algorithm can efficiently produce posterior draws of $\mathbf{B}$ conditional on linear inequality restrictions, the additional accept-reject steps are specifically implemented to handle the nonlinear inequality restrictions. This approach differs from the conventional accept-reject method, which is designed to address all linear and nonlinear inequality restrictions together in a single step. Imposing inequality restrictions, whether linear or nonlinear, often truncates the support of the posterior distribution into a region with a small probability. This can lead to an extremely low acceptance rate for the conventional accept-reject approach, particularly when the total number of inequality restrictions is large. However, by considering only a relatively small number of nonlinear inequality restrictions in the accept-reject steps, we can significantly reduce the computational cost compared to the conventional accept-reject approach.

2.5.3. Alternative priors

While we describe our algorithm under an independent truncated Gaussian prior for $\mathbf{B}$, it is clear that the proposed method is applicable to a class of priors that can be represented as a scale mixture of Gaussian distributions. This class of priors includes the t-distributed prior (Baumeister and Hamilton, 2015, 2019), stochastic search variable selection prior (George et al., 2008), Horseshoe prior (Carvalho et al., 2010), Normal-Gamma prior (Brown and Griffin, 2010), and Dirichlet–Laplace prior (Bhattacharya et al., 2015).

Our proposed methodology, with some modifications, can also be used to consider models with the Normal-Generalized-Normal (NGN) prior proposed by Arias et al. (2018, 2021). The NGN prior is a conjugate prior that has been widely used in the literature, which is typically imposed on the structural parameters for SVARs or Proxy-SVARs represented in the form of A-model specification (an overview of the Proxy-SVAR is provided in Section 3). In Appendix B, we provide full details about how our proposed method can be used for inference in Proxy-SVARs when the NGN prior is considered.

2.5.4. Exact sampling for w_1

A practical concern when applying our algorithm may arise in the step of drawing w_1. Recall that both Algorithm 1 and Algorithm 2 make use of the method proposed in Proposition 2 to obtain a proposal draw $w_1^* \sim q^*(w_1^*)$, which requires drawing from an absolute normal distribution $\mathcal{AN}(a, b)$ with $a = -\frac{\gamma_1}{2\sqrt{\gamma_2(T-2)}}$ and $b = \frac{1}{T-2}$. As discussed in Section 2.3, drawing from $\mathcal{AN}(a, b)$ can be achieved by implementing the normal mixture approximation approach developed by Villani (2009). This approximation often works well when b is large (see also the discussion in Appendix C of Villani (2009)). However, the absolute normal distribution of our interest has $b = \frac{1}{T-2}$. When the sample size T is large, this leads to a small value of b, which might in turn lead to poor approximation accuracy. In

[4] In MATLAB, one way to obtain $\widetilde{\mathbf{R}}_i^E$ and $\widetilde{\mathbf{d}}_i$ given $\bar{\mathbf{R}}_i$ and $\bar{\mathbf{d}}_i$ is to implement the following steps: (1) compute the QR decomposition of $\bar{\mathbf{R}}_i$, i.e., [Q,R] = qr($\bar{\mathbf{R}}_i$); (2) set $\widetilde{\mathbf{R}}_i^E$ to be the first $\widetilde{r}_i^E$ rows of R, i.e., $\widetilde{\mathbf{R}}_i^E$ = R(1: $\widetilde{r}_i^E$, :); (3) set $\widetilde{\mathbf{d}}_i$ to be the first $\widetilde{r}_i^E$ elements of Q'$\bar{\mathbf{d}}_i$, i.e., $\widetilde{\mathbf{d}}_i$ = Q(:, 1: $\widetilde{r}_i^E$)'$\bar{\mathbf{d}}_i$.

[5] Specifically, both the step of drawing $\mathbf{B}$ and the step of drawing the autoregressive coefficients need to include an additional accept-reject step.

Appendix C, we develop an alternative approach that enables exact sampling from the conditional posterior $p(w_1|\mathbf{w}_{-1}, \mathbf{B}_{-1}, \widetilde{\mathbf{y}})$. In the application of this paper, we find that the exact sampling approach developed in Appendix C and the normal mixture approximation approach of Villani (2009) produce almost identical results.

2.5.5. Limitations

One limitation of our proposed method is that it cannot handle cases where equality restrictions are imposed on nonlinear functions of the SVAR parameters, such as zero restrictions on impulse response functions at higher horizons, historical decompositions, and the reliability condition of proxy variables in Proxy-SVAR analysis.[6] For instance, when equality restrictions are simultaneously imposed on matrices $\mathbf{B}$ and $\mathbf{B}^{-1}$, such as the zero restrictions used in the empirical study of Mertens and Montiel Olea (2018). If such equality restrictions are of interest, one should use the methods proposed by Arias et al. (2018, 2021) and Antolín-Díaz and Rubio-Ramírez (2018). One appealing property of the methods proposed by Arias et al. (2018, 2021) is that their algorithms allow for inference based on a prior that assigns the same density to observationally equivalent structural parameters.[7] In contrast, the methodology proposed in this paper is not applicable to such a prior. Whether our proposed methodology can be extended to accommodate a more general form of nonlinear equality restrictions and a wider class of prior distributions remains an important research question, and we will leave it for future work.

3. Empirical model: Proxy-SVAR

The empirical application of this paper is based on a Proxy-SVAR, which is one of the most popular variants of SVARs with equality and inequality restrictions. In this section, we first provide an overview of the Proxy-SVAR. We then point out some key differences between our proposed methodology and the existing methods, and highlight the advantage of our methodology when multiple proxy variables are available for identifying multiple structural shocks.

Let $\boldsymbol{\zeta}_t$ be a $\widetilde{n} \times 1$ vector of endogenous variables and consider the following SVAR:

$$\boldsymbol{\zeta}_t = \mathbf{c}_\zeta + \boldsymbol{\Gamma}_1 \boldsymbol{\zeta}_{t-1} + \cdots + \boldsymbol{\Gamma}_p \boldsymbol{\zeta}_{t-p} + \boldsymbol{\Gamma}_0 \boldsymbol{\eta}_t, \quad \boldsymbol{\eta}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{\widetilde{n}}), \tag{24}$$

where $\widetilde{\mathbf{c}}_\zeta$ is an $\widetilde{n} \times 1$ vector of intercepts, $\boldsymbol{\Gamma}_i$ is $\widetilde{n} \times \widetilde{n}$ for $i = 0, \ldots, p$, with $\boldsymbol{\Gamma}_0$ invertible, and $\boldsymbol{\eta}_t = (\eta_{1,t}, \ldots, \eta_{\widetilde{n},t})'$ is a $\widetilde{n} \times 1$ vector of structural shocks. As is well known, without additional information, the structural shocks $\boldsymbol{\eta}_t$ are not identified. In a Proxy-SVAR setting, we assume that there is a $k \times 1$ vector of proxy variables $\mathbf{m}_t = (m_{1,t}, \ldots, m_{k,t})'$, containing external information that can be exploited to identify a subset of the structural shocks in $\boldsymbol{\eta}_t$. We assume that the proxy variables $\mathbf{m}_t$ have the following representation:

$$\mathbf{m}_t = \mathbf{c}_m + \sum_{i=1}^{p} \left(\boldsymbol{\Phi}_i \boldsymbol{\zeta}_{t-i} + \boldsymbol{\Psi}_i \mathbf{m}_{t-i} \right) + \boldsymbol{\Phi}_0 \boldsymbol{\eta}_t + \boldsymbol{\Psi}_0 \boldsymbol{\upsilon}_t, \quad \boldsymbol{\upsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_k), \tag{25}$$

where $\mathbf{c}_m$ is a $k \times 1$ vector of intercepts, $\boldsymbol{\Phi}_i$ is $k \times \widetilde{n}$ and $\boldsymbol{\Psi}_i$ is $k \times k$ for $i = 0, \ldots, p$, and $\boldsymbol{\upsilon}_t$ is $k \times 1$ that is assumed to be independent of $\boldsymbol{\eta}_t$.

If we define an $n \times 1$ vector $\mathbf{y}_t = (\boldsymbol{\zeta}_t', \mathbf{m}_t')'$ with $n = \widetilde{n} + k$, then it follows that we can represent (24) and (25) as an augmented SVAR:

$$\mathbf{y}_t = \mathbf{c} + \mathbf{A}_1 \mathbf{y}_{t-1} + \cdots + \mathbf{A}_p \mathbf{y}_{t-p} + \mathbf{B} \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n), \tag{26}$$

where $\mathbf{c} = (\mathbf{c}_\zeta', \mathbf{c}_m')'$, $\boldsymbol{\epsilon}_t = (\boldsymbol{\eta}_t', \boldsymbol{\upsilon}_t')'$, and the parameter matrices are given by

$$\mathbf{B} = \begin{pmatrix} \boldsymbol{\Gamma}_0 & \mathbf{0}_{\widetilde{n} \times k} \\ \boldsymbol{\Phi}_0 & \boldsymbol{\Psi}_0 \end{pmatrix}, \quad \mathbf{A}_i = \begin{pmatrix} \boldsymbol{\Gamma}_i & \mathbf{0}_{\widetilde{n} \times k} \\ \boldsymbol{\Phi}_i & \boldsymbol{\Psi}_i \end{pmatrix}, \quad i = 1, \ldots, p. \tag{27}$$

The augmented SVAR (26) is a completed version of the simplified model specified in (1) that incorporates the autoregressive conditional mean component. The zero restrictions in (27) implies that $\mathbf{m}_t$ and $\boldsymbol{\upsilon}_t$ contain no predictive information for the endogenous variables $\boldsymbol{\zeta}_t$. Let $\mathbf{A}^+ = (\mathbf{c}, \mathbf{A}_1, \ldots, \mathbf{A}_p)$ and $\mathbf{X}_t = (1, \mathbf{y}_1', \ldots, \mathbf{y}_p')'$, then we can write (26) more compactly as

$$\mathbf{y}_t = \mathbf{A}^+ \mathbf{X}_t + \mathbf{B} \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n). \tag{28}$$

Valid proxy variables used to identify a set of structural shocks in $\boldsymbol{\eta}_t$ are required to be correlated with the structural shocks of interest and uncorrelated with the other remaining structural shocks. To be specific, we partition the $\widetilde{n} \times 1$ structural shocks as $\boldsymbol{\eta}_t = (\boldsymbol{\eta}_t^{1'}, \boldsymbol{\eta}_t^{2'})'$, where $\boldsymbol{\eta}_t^1$ is $\widetilde{n}_1 \times 1$, $\boldsymbol{\eta}_t^2$ is $\widetilde{n}_2 \times 1$ and $\widetilde{n} = \widetilde{n}_1 + \widetilde{n}_2$. Without loss of generality, suppose that our primary interest is to identify the first $\widetilde{n}_1$ structural shocks in $\boldsymbol{\eta}_t$, i.e., $\boldsymbol{\eta}_t^1$, using the k proxy variables in $\mathbf{m}_t$, and we assume $\widetilde{n}_1 \leq k$. The proxy variables $\mathbf{m}_t$ need to satisfy the following relevant and exogeneity conditions:

$$\text{Relevance condition: } \mathbb{E}(\mathbf{m}_t \boldsymbol{\eta}_t^{1'}) = \boldsymbol{\Omega}, \ \operatorname{rk}(\boldsymbol{\Omega}) = \widetilde{n}_1; \tag{29}$$

[6] This is because such equality restrictions also require imposing nonlinear restrictions on the autoregressive coefficients of the model, which is beyond the scope of this paper.

[7] Furthermore, the methods proposed by Arias et al. (2018, 2021) can handle a broader class of prior distributions for observationally equivalent objects of interest, such as the Uniform prior distribution (Arias et al., 2023).

$$\text{Exogeneity condition: } \mathbb{E}(\mathbf{m}_t \boldsymbol{\eta}_t^{2\prime}) = \mathbf{0}_{k\times\widetilde{n}_2}. \tag{30}$$

If the relevance and exogeneity conditions hold, we have

$$\mathbb{E}(\mathbf{m}_t \boldsymbol{\eta}_t') = \left(\mathbb{E}(\mathbf{m}_t \boldsymbol{\eta}_t^{1\prime}), \mathbb{E}(\mathbf{m}_t \boldsymbol{\eta}_t^{2\prime})\right) = \left(\boldsymbol{\Omega}, \mathbf{0}_{k\times\widetilde{n}_2}\right). \tag{31}$$

Let $\mathbf{J}_1 = (\mathbf{0}_{k\times\widetilde{n}}, \mathbf{I}_{k\times k})$ and $\mathbf{J}_2 = (\mathbf{I}_{\widetilde{n}\times\widetilde{n}}, \mathbf{0}_{\widetilde{n}\times k})$, then we can express $\mathbf{m}_t$ and $\boldsymbol{\eta}_t$ as

$$\mathbf{m}_t = \mathbf{J}_1\mathbf{y}_t = \mathbf{J}_1\mathbf{A}^+\mathbf{X}_t + \mathbf{J}_1\mathbf{B}\boldsymbol{\epsilon}_t \quad \text{and} \quad \boldsymbol{\eta}_t = \mathbf{J}_2\boldsymbol{\epsilon}_t,$$

which gives

$$\mathbb{E}(\mathbf{m}_t\boldsymbol{\eta}_t') = \mathbb{E}(\mathbf{J}_1\mathbf{A}^+\mathbf{X}_t\boldsymbol{\epsilon}_t'\mathbf{J}_2') + \mathbb{E}(\mathbf{J}_1\mathbf{B}\boldsymbol{\epsilon}_t\boldsymbol{\epsilon}_t'\mathbf{J}_2') = \mathbb{E}(\mathbf{J}_1\mathbf{B}\boldsymbol{\epsilon}_t\boldsymbol{\epsilon}_t'\mathbf{J}_2') = \mathbf{J}_1\mathbf{B}\mathbf{J}_2'. \tag{32}$$

From the results given in (27), (31) and (32), we obtain another set of zero restrictions for $\mathbf{B}$:

$$\mathbf{J}_1\mathbf{B}\mathbf{J}_2' = \left(\boldsymbol{\Omega}, \mathbf{0}_{k\times\widetilde{n}_2}\right) = \boldsymbol{\Phi}_0. \tag{33}$$

In summary, a Proxy-SVAR can be represented as an augmented SVAR (26) conditional on the zero restrictions given in (27) and (33). However, these zero restrictions are not sufficient for uniquely identifying $\mathbf{B}$. Following the discussions in Arias et al. (2021), it can be shown that $\widehat{\mathbf{B}} = \mathbf{B}\mathbf{Q}$ and $\mathbf{B}$ are observationally equivalent if and only if $\mathbf{Q} = \text{diag}(\mathbf{Q}_1, \mathbf{Q}_2, \mathbf{Q}_3)$, where $\mathbf{Q}_1 \in \mathcal{O}(\widetilde{n}_1)$, $\mathbf{Q}_2 \in \mathcal{O}(\widetilde{n}_2)$, $\mathbf{Q}_3 \in \mathcal{O}(k)$, and $\mathcal{O}(q)$ denotes the set of all $q \times q$ orthogonal matrices. As pointed out by Arias et al. (2021), imposing additional sign restrictions on the free parameters in $\mathbf{B}$ can provide further identifying information. For example, if one proxy variable is available for identifying one structural shock, i.e., $k = \widetilde{n}_1 = 1$, then the first column of $\mathbf{B}$, i.e., $\mathbf{b}_1$, is identified up to sign changes because $\mathbf{Q}_1$ can either be 1 or -1. Hence, $\mathbf{b}_1$ is point-identified if we impose a sign restriction on one of its free parameters.

The above discussions about the Proxy-SVAR are similar to Arias et al. (2021). The only difference is that their econometric framework is based on the structural parameters $\bar{\mathbf{A}}^+ = \mathbf{B}^{-1}\mathbf{A}^+$ and $\bar{\mathbf{B}} = \mathbf{B}^{-1}$, while ours is based on $\mathbf{A}^+$ and $\mathbf{B}$. One difficulty in conducting Bayesian inference on the structural parameters $(\bar{\mathbf{A}}^+, \bar{\mathbf{B}})$ is that the zero restrictions given in (27) and (33) induce nonlinear restrictions on the parameter space of $(\bar{\mathbf{A}}^+, \bar{\mathbf{B}})$. These nonlinear restrictions often pose challenges in posterior computation, especially when additional sign restrictions are imposed on $(\mathbf{A}^+, \mathbf{B})$. In our application, we consider an independent Gaussian prior over the parameters $(\mathbf{A}^+, \mathbf{B})$.[8] Therefore, given $\mathbf{A}^+$, the algorithm developed in Section 2 can be applied to draw $\mathbf{B}$ subject to linear equality and inequality restrictions in the form of (2) and (3), which include the zero and sign restrictions commonly used in many Proxy-SVAR studies. On the other hand, using independent prior for $(\mathbf{A}^+, \mathbf{B})$ allows us to impose a variety of flexible shrinkage prior beliefs on the structural parameter $\mathbf{A}^+$, which have been shown to be beneficial in overcoming the overparameterization concerns and producing sensible results for structural analysis under the setting of large models. Since the step for drawing $\mathbf{A}^+$ is straightforward and similar to those used elsewhere in the literature, we defer the details of the prior and the sampling procedure for $\mathbf{A}^+$ to Section S.1 of the Supplementary Material.

One advantage of our proposed methodology over the existing methods for Proxy-SVAR analysis is that our algorithm can handle various cases where multiple proxy variables are used to identify multiple structural shocks, allowing the number of proxy variables to differ from the number of structural shocks. For instance, using two proxy variables to identify one structural shock. It is important to note that different cases of using multiple proxy variables to identify multiple structural shocks lead to different patterns of zero restrictions on $\mathbf{B}$. As we can see in Section 2, our proposed algorithm is capable of imposing any pattern of zero restrictions on $\mathbf{B}$ as long as the restricted $\mathbf{B}$ remains invertible.

Another advantage of the proposed methodology is that it allows us to consider Proxy-SVARs with over-identifying restrictions. This is a case that cannot be addressed by the existing methods, such as Arias et al. (2021) and Braun and Brüggemann (2023). An over-identified Proxy-SVAR is often of interest in situations where researchers strongly believe that a subset of the proxy variables in $\mathbf{m}_t$ is only correlated with a subset of the structural shocks in $\boldsymbol{\eta}_t^1$, but uncorrelated with all the other remaining structural shocks to be identified. This requires imposing additional zero restrictions on $\boldsymbol{\Omega}$. Moreover, it turns out that such over-identifying zero restrictions would provide more information for identifying multiple structural shocks. To illustrate this, suppose that the proxy variables $\mathbf{m}_t$ and the structural shocks of interest $\boldsymbol{\eta}_t^1$ can be divided into g groups, written as $\mathbf{m}_t = (\mathbf{m}_{1,t}', \ldots, \mathbf{m}_{g,t}')'$ and $\boldsymbol{\eta}_t^1 = (\boldsymbol{\eta}_{1,t}^{1\prime}, \ldots, \boldsymbol{\eta}_{g,t}^{1\prime})'$, where $\mathbf{m}_{i,t}$ is $k_i \times 1$, $\boldsymbol{\eta}_{i,t}^1$ is $\widetilde{n}_{1,i} \times 1$ and $k_i \le \widetilde{n}_{1,i}$, $i = 1, \ldots, g$, with $\sum_{i=1}^g k_i = k$ and $\sum_{i=1}^g \widetilde{n}_{1,i} = \widetilde{n}_1$. If $\mathbf{m}_{i,t}$ is assumed to be correlated with $\boldsymbol{\eta}_{i,t}^1$ and uncorrelated with $\boldsymbol{\eta}_{j,t}^1$ for $i = 1, \ldots, g$ and $i \ne j$, that is, $\mathbb{E}(\mathbf{m}_{i,t}\boldsymbol{\eta}_{i,t}^{1\prime}) = \boldsymbol{\Omega}_i$ with $\text{rk}(\boldsymbol{\Omega}_i) = \widetilde{n}_{1,i}$ and $\mathbb{E}(\mathbf{m}_{i,t}\boldsymbol{\eta}_{j,t}^{1\prime}) = \mathbf{0}$ for $i = 1, \ldots, g$

[8] In general, we can write the prior for $(\mathbf{A}_+, \mathbf{B})$ as $\text{vec}(\mathbf{A}_+) \sim \mathcal{N}(\boldsymbol{\mu}_a, \mathbf{V}_a)$ and $\text{vec}(\mathbf{B}) \sim \mathcal{N}(\boldsymbol{\mu}_b, \mathbf{V}_b)$, which has the density function:

$$p(\mathbf{A}_+, \mathbf{B}) \propto e^{-\frac{1}{2}(\text{vec}(\mathbf{A}_+)-\boldsymbol{\mu}_a)'\mathbf{V}_a^{-1}(\text{vec}(\mathbf{A}_+)-\boldsymbol{\mu}_a)} e^{-\frac{1}{2}(\text{vec}(\mathbf{B})-\boldsymbol{\mu}_b)'\mathbf{V}_b^{-1}(\text{vec}(\mathbf{B})-\boldsymbol{\mu}_b)}.$$

Using a change of variable technique with $(\bar{\mathbf{A}}_+, \bar{\mathbf{B}}) = (\mathbf{B}^{-1}\mathbf{A}_+, \mathbf{B}^{-1})$ yields the Jacobian of transformation $|\det(\bar{\mathbf{B}})|^{-(2n+np+1)}$ and the following prior density function for the structural parameters $(\bar{\mathbf{A}}_+, \bar{\mathbf{B}})$:

$$p(\bar{\mathbf{A}}_+, \bar{\mathbf{B}}) \propto |\det(\bar{\mathbf{B}})|^{-(2n+np+1)} e^{-\frac{1}{2}[\text{vec}(\bar{\mathbf{A}}_+)-(\mathbf{I}_{np+1}\otimes\bar{\mathbf{B}})\boldsymbol{\mu}_a]'(\mathbf{I}_{np+1}\otimes\bar{\mathbf{B}}^{-1})'\mathbf{V}_a^{-1}(\mathbf{I}_{np+1}\otimes\bar{\mathbf{B}}^{-1})[\text{vec}(\bar{\mathbf{A}}_+)-(\mathbf{I}_{np+1}\otimes\bar{\mathbf{B}})\boldsymbol{\mu}_a]}$$
$$\times\, e^{-\frac{1}{2}(\text{vec}(\bar{\mathbf{B}}^{-1})-\boldsymbol{\mu}_b)'\mathbf{V}_b^{-1}(\text{vec}(\bar{\mathbf{B}}^{-1})-\boldsymbol{\mu}_b)}.$$

This shows that the prior of $\text{vec}(\bar{\mathbf{A}}_+)$ conditional on $\bar{\mathbf{B}}$ is a Gaussian distribution $\left(\text{vec}(\bar{\mathbf{A}}_+)|\bar{\mathbf{B}}\right) \sim \mathcal{N}\left((\mathbf{I}_{np+1}\otimes\bar{\mathbf{B}})\boldsymbol{\mu}_a, (\mathbf{I}_{np+1}\otimes\bar{\mathbf{B}})\mathbf{V}_a(\mathbf{I}_{np+1}\otimes\bar{\mathbf{B}})'\right)$ and the marginal prior of $\bar{\mathbf{B}}$ has a density function $p(\bar{\mathbf{B}}) \propto |\det(\bar{\mathbf{B}})|^{-(2n+np+1)} e^{-\frac{1}{2}(\text{vec}(\bar{\mathbf{B}}^{-1})-\boldsymbol{\mu}_b)'\mathbf{V}_b^{-1}(\text{vec}(\bar{\mathbf{B}}^{-1})-\boldsymbol{\mu}_b)}$. For a Proxy-SVAR, the implied prior for $(\bar{\mathbf{A}}_+, \bar{\mathbf{B}})$ needs to be restricted to satisfy the zero restrictions given in (27) and (33), where $(\mathbf{A}_+, \mathbf{B}) = (\bar{\mathbf{B}}^{-1}\bar{\mathbf{A}}_+, \bar{\mathbf{B}}^{-1})$.

and $i \neq j$, then we have $\boldsymbol{\Omega} = \text{diag}(\boldsymbol{\Omega}_1, \dots, \boldsymbol{\Omega}_g)$. With the relevance condition (29) and exogeneity condition (30), we can show that $\widehat{\mathbf{B}} = \mathbf{B}\mathbf{Q}$ is observationally equivalent to $\mathbf{B}$ if and only if $\mathbf{Q} = \text{diag}(\mathbf{Q}_1, \mathbf{Q}_2, \mathbf{Q}_3)$, where $\mathbf{Q}_1 = \text{diag}(\mathbf{Q}_{1,1}, \dots, \mathbf{Q}_{1,g})$, $\mathbf{Q}_{1,i} \in \mathcal{O}(\widetilde{n}_{1,i})$ for $i = 1, \dots, g$, $\mathbf{Q}_2 \in \mathcal{O}(\widetilde{n}_2)$ and $\mathbf{Q}_3 \in \mathcal{O}(k)$. In particular, if each group of structural shocks of interest contains only one structural shock, i.e., $\widetilde{n}_{1,i} = 1$ for $i = 1, \dots, g$, then we have $g = \widetilde{n}_1$ and $\mathbf{Q}_{1,i}$ can either be 1 or -1 for $i = 1, \dots, g$. As a result, the first $\widetilde{n}_1$ columns of $\mathbf{B}$, i.e., $\mathbf{b}_1, \dots, \mathbf{b}_{\widetilde{n}_1}$, are uniquely identified up to sign changes. In this case, for $i = 1, \dots, \widetilde{n}_1$, $\mathbf{b}_i$ is point-identified if a sign restriction is imposed on one of its free parameters.

4. Empirical application: Contributions of macroeconomic shocks to the 2007–09 recession

This section revisits the empirical analysis in Stock and Watson (2012) that investigates the contributions of macroeconomic shocks to the fluctuations in GDP growth and employment during the 2007–09 recession. We employ a large Proxy-SVAR with multiple proxy variables to identify multiple macroeconomic structural shocks. More specifically, we focus on assessing the role of 5 macroeconomic shocks — oil, monetary policy, productivity, liquidity and financial risk, and fiscal policy shocks — in explaining the fluctuations in GDP growth and employment during the 2007–09 recession period.[9] The major difference between our analysis and the one in Stock and Watson (2012) is that they use a dynamic factor model with one proxy variable to identify one structural shock of interest at a time, while we use a large Proxy-SVAR that provides a unified econometric framework for simultaneously identifying all the structural shocks of interest.

We consider the Minnesota-type adaptive hierarchical prior (Cross et al., 2020; Chan, 2021) for the autoregressive coefficients in the VAR Eqs. (24). This prior allows for asymmetric shrinkage effects on the coefficients associated with own lags and lags of other endogenous variables, while treating the hyperparameters governing the degree of shrinkage as unknown parameters to be estimated in a databased fashion.[10] Full details of the prior and the sampling procedures are provided in Section S.1 of the Supplementary Material. All estimation results reported in our empirical application are based on 5000 posterior draws by retaining every 20th draws from a total of 100 000 draws after discarding a burn-in period of 5000 draws. We evaluate the mixing properties and correctness of the MCMC sampler in Section S.2 of the Supplementary Material, which shows that our MCMC sampler is correctly implemented and performs well in producing posterior draws with low autocorrelations.

4.1. Data and model specification

Our application uses a dataset that consists of 30 quarterly US macroeconomic variables with a sample period from 1987Q1 to 2011Q2. The data are first transformed to be stationary and then standardized to have unit variances. We use 9 quarterly proxy variables constructed in the relevant studies in the literature to identify the 5 structural shocks of interest. All data used in our empirical study can be obtained from the online Appendix of Stock and Watson (2012), and we refer readers to their paper for more details. The complete list of the macroeconomic variables and the proxy variables are given in Section S.3 of the Supplementary Material.

The Proxy-SVAR considered in our empirical analysis includes $k = 9$ proxy variables that are used to simultaneously identify $\widetilde{n}_1 = 5$ structural shocks. Using the notations introduced in Section 3, the vector of structural shocks to be identified is denoted as $\boldsymbol{\eta}_t^1 = (\eta_{1,t}, \eta_{2,t}, \eta_{3,t}, \eta_{4,t}, \eta_{5,t})'$, where $\eta_{1,t}$, $\eta_{2,t}$, $\eta_{3,t}$, $\eta_{4,t}$ and $\eta_{5,t}$ represent oil, monetary policy, productivity, liquidity and financial risk, and fiscal shocks respectively. The vector of proxy variables is denoted as $\mathbf{m}_t = (m_{1,t}, m_{2,t}, m_{3,t}, m_{4,t}, m_{5,t}, m_{6,t}, m_{7,t}, m_{8,t}, m_{9,t})'$, where $m_{i,t}$, for $i = 1, \dots, 9$, are defined in Section S.3 of the Supplementary Material. We let $\mathbf{c}_m = \mathbf{0}$ and $\boldsymbol{\Psi}_i = \mathbf{0}$, $i = 1, \dots, p$. Under this setting and given the zero restrictions in (33), the equations for the proxy variables (25) reduce to

$$\mathbf{m}_t = \sum_{i=1}^{p} \boldsymbol{\Phi}_i \mathbf{m}_{t-i} + \boldsymbol{\Omega}\boldsymbol{\eta}_{1,t} + \boldsymbol{\Psi}_0 \boldsymbol{v}_t, \quad \boldsymbol{v}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_k). \tag{34}$$

We impose additional zero restrictions on $\boldsymbol{\Omega}$, which is of dimension 9×5, such that each proxy variable is only correlated with the structural shock it is designed to identify and uncorrelated with all the other remaining structural shocks. Specifically, following the discussion in Section 3, we divide the proxy variables in $\mathbf{m}_t$ into $g = 5$ groups and write $\mathbf{m}_t = (\mathbf{m}_{1,t}', \dots, \mathbf{m}_{5,t}')'$, where $\mathbf{m}_{1,t} = m_{1,t}$, $\mathbf{m}_{2,t} = (m_{2,t}, m_{3,t})'$, $\mathbf{m}_{3,t} = (m_{4,t}, m_{5,t})'$, $\mathbf{m}_{4,t} = (m_{6,t}, m_{7,t})'$ and $\mathbf{m}_{5,t} = (m_{8,t}, m_{9,t})'$, which gives $k_1 = 1$ and $k_i = 2$ for $i = 2, \dots, 5$. We assume that $\boldsymbol{\Omega}_i = \mathbb{E}(\mathbf{m}_{i,t}\eta_{i,t})$ is a $k_i \times 1$ nonzero vector and $\mathbb{E}(\mathbf{m}_{i,t}\eta_{j,t}) = \mathbf{0}$, for $i = 1 \dots, 5$, $i \neq j$. As a result, we have $\boldsymbol{\Omega} = \text{diag}(\boldsymbol{\Omega}_1, \dots, \boldsymbol{\Omega}_5)$. With this setting, we have shown in Section 3 that the structural shocks of interest $\boldsymbol{\eta}_t^1 = (\eta_{1,t}, \eta_{2,t}, \eta_{3,t}, \eta_{4,t}, \eta_{5,t})'$ are uniquely identified up to sign changes. In other words, the first 5 columns of $\mathbf{B}$, i.e., $\mathbf{b}_1, \dots, \mathbf{b}_5$, are uniquely identified up to sign changes. We set $\boldsymbol{\Psi}_0$ and $\boldsymbol{\Phi}_i$, $i = 1, \dots, p$, to be diagonal matrices, implying that the proxy variables are assumed to follow independent AR(p) processes, and the lag length of the model is set to $p = 4$.

[9] Besides these 5 structural shocks, Stock and Watson (2012) also consider uncertainty shocks. However, as also mentioned by them, uncertainty shocks and liquidity and financial risk shocks are highly correlated, which gives rise to a difficulty in the interpretation of the effects of these two structural shocks. Thus, we only consider liquidity and financial risk shocks in our analysis and do not attempt to separately identify liquidity and financial risk shocks and uncertainty shocks.

[10] As pointed out by Kilian and Lütkepohl (2017) (see Chapter 17), using the conventional shrinkage prior for estimating large VARs might limit the degrees of freedom of the model and distort the parameter estimates. This is because the hyperparameters of the shrinkage prior are often fixed at specific small values, resulting in small prior variances for the autoregressive coefficients. To address this concern, we assume an uninformative prior (see Section S.1 of the Supplementary Material) for the hyperparameters. Therefore, we do not constrain the hyperparameters to small values a priori, but allow them to be estimated from the data.

Table 1
Competing models in the model comparison exercise.

PSVAR-30	$\boldsymbol{\Omega} = \text{diag}(\boldsymbol{\Omega}_1, \dots, \boldsymbol{\Omega}_5)$ and $\boldsymbol{\Phi}_i$ is a diagonal matrix for $i = 1, \dots, p$
PSVAR-30-A1	$\boldsymbol{\Omega} = \text{diag}(\boldsymbol{\Omega}_1, \dots, \boldsymbol{\Omega}_5)$ and $\boldsymbol{\Phi}_i = \mathbf{0}$ for $i = 1, \dots, p$
PSVAR-30-A2	$\boldsymbol{\Omega}$ is unrestricted and $\boldsymbol{\Phi}_i$ is a diagonal matrix for $i = 1, \dots, p$
PSVAR-30-A3	$\boldsymbol{\Omega}$ is unrestricted and $\boldsymbol{\Phi}_i = \mathbf{0}$ for $i = 1, \dots, p$

Table 2
Estimated log marginal likelihood (log ML).

	PSVAR-30	PSVAR-30-A1	PSVAR-30-A2	PSVAR-30-A3
log ML	−3906	−3948	−3949	−3980

In our application, we do not exploit additional sign restrictions to point-identify each of the 5 structural shocks. This is because our empirical analysis is based on the historical and forecast error variance decompositions attributed to the structural shocks of interest (see Section 4.3), which are point-identified without requiring further information. For example, consider the first column of $\mathbf{B}$, i.e., $\mathbf{b}_1$, which represents the contemporaneous impacts of the oil shock η_t^O. The historical and forecast error variance decompositions related to the oil shocks depend solely on the product $\mathbf{b}_1\eta_t^O$. Since both $\mathbf{b}_1$ and η_t^O are identified subject to sign changes, the product $\mathbf{b}_1\eta_t^O$ is always point-identified.[11]

The proxy variables considered in our application contain many missing observations within the sample period. To address this issue, we treat the missing observations as parameters to be estimated. Due to the modular nature of MCMC methods, it is straightforward to incorporate an additional step in our proposed algorithm to draw these missing observations conditional on the model parameters. Since the Proxy-SVAR is a Gaussian state space model, drawing from the conditional distribution of the missing observations can be accomplished using the algorithm developed by Chan et al. (2023). This algorithm is built upon the precision-based sampling method (Rue, 2001; Chan and Jeliazkov, 2009; McCausland et al., 2011) and it enables the efficient drawing of all missing observations in the proxy variables in one step.

4.2. Testing for exogeneity of proxy variables

As discussed in the previous section, the 30-variable Proxy-SVAR, denoted as PSVAR-30, used in our application is characterized by two important features: (1) the proxy variables are assumed to follow independent AR(p) processes, i.e., $\boldsymbol{\Phi}_i$ are diagonal matrices for $i = 1, \dots, p$; (2) the correlations between the proxy variables are restricted such that each proxy variable is correlated only with a specific structural shock of interest and uncorrelated with the other structural shocks, i.e., $\boldsymbol{\Omega} = \text{diag}(\boldsymbol{\Omega}_1, \dots, \boldsymbol{\Omega}_5)$. In this section, we conduct a formal model comparison exercise to check whether these two modeling features are supported by the data.

To this end, we compare our PSVAR-30 with three alternative models. The first model, denoted as PSVAR-30-A1, is a restricted version of our PSVAR-30 in which the proxy variables are specified to have zero conditional means instead of being modeled as independent AR(p) processes, that is, we set $\boldsymbol{\Phi}_i = \mathbf{0}$, $i = 1, \dots, p$. The second model is denoted as PSVAR-30-A2, which is specified as the same as the PSVAR-30, except that we allow all proxy variables to be correlated with all structural shocks of interest by setting $\boldsymbol{\Omega}$ to be an unrestricted matrix. The third model we consider assumes that $\boldsymbol{\Omega} = \text{diag}(\boldsymbol{\Omega}_1, \dots, \boldsymbol{\Omega}_5)$ and $\boldsymbol{\Phi}_i = \mathbf{0}$, $i = 1, \dots, p$, and we denote it as PSVAR-30-A3. Table 1 provides a summary of the competing models in our model comparison exercise.

A natural measure for Bayesian model comparison is the marginal likelihood, which is defined for a given model as

$$p(\mathbf{y}^o) = \int p(\mathbf{y}^o|\boldsymbol{\theta})p(\boldsymbol{\theta})d\boldsymbol{\theta},$$

where $\mathbf{y}^o = (\mathbf{y}_1^{o\prime}, \dots, \mathbf{y}_T^{o\prime})'$ is a vector of observed data, $\boldsymbol{\theta}$ is a vector collecting the model-specific parameters, and $p(\mathbf{y}^o|\boldsymbol{\theta})$ and $p(\boldsymbol{\theta})$ denote the likelihood function and the prior density for the model. The marginal likelihood can be interpreted as the joint marginal density evaluated at the observed data under the prior. A large value of the marginal likelihood indicates strong statistical evidence provided by the data in favor of the model. The marginal likelihoods are estimated using the modified harmonic mean estimator of Gelfand and Dey (1994), and the details of the estimation are provided in Section S.4 of the Supplementary Material.

Table 2 reports the log marginal likelihood estimates. Two main findings can be drawn from these results. First, our results show that the differences in log marginal likelihoods between the PSVAR-30 and the alternative models are all larger than 40, highlighting overwhelming evidence in favor of our PSVAR-30. Second, it is apparent that models specifying independent AR(p) processes for the proxy variables or incorporating a restricted $\boldsymbol{\Omega} = \text{diag}(\boldsymbol{\Omega}_1, \dots, \boldsymbol{\Omega}_5)$ perform significantly better than their counterparts without these two features. For instance, the log marginal likelihoods for both PSVAR-30-A1 and PSVAR-30-A2 are significantly higher than that of the PSVAR-30-A3, providing strong evidence supporting the independent AR(p) specification for the proxy variables and the restricted correlation structure between the proxy variables and the structural shocks of interest.

[11] Baumeister and Hamilton (2015) show that prior distribution can have a non-negligible effect on posterior inference for set-identified estimates, even when the sample size is large. However, our empirical results are based on the historical and forecast error variance decompositions of the identified shocks, which are point-identified and, therefore, do not suffer from the criticism raised by Baumeister and Hamilton (2015).

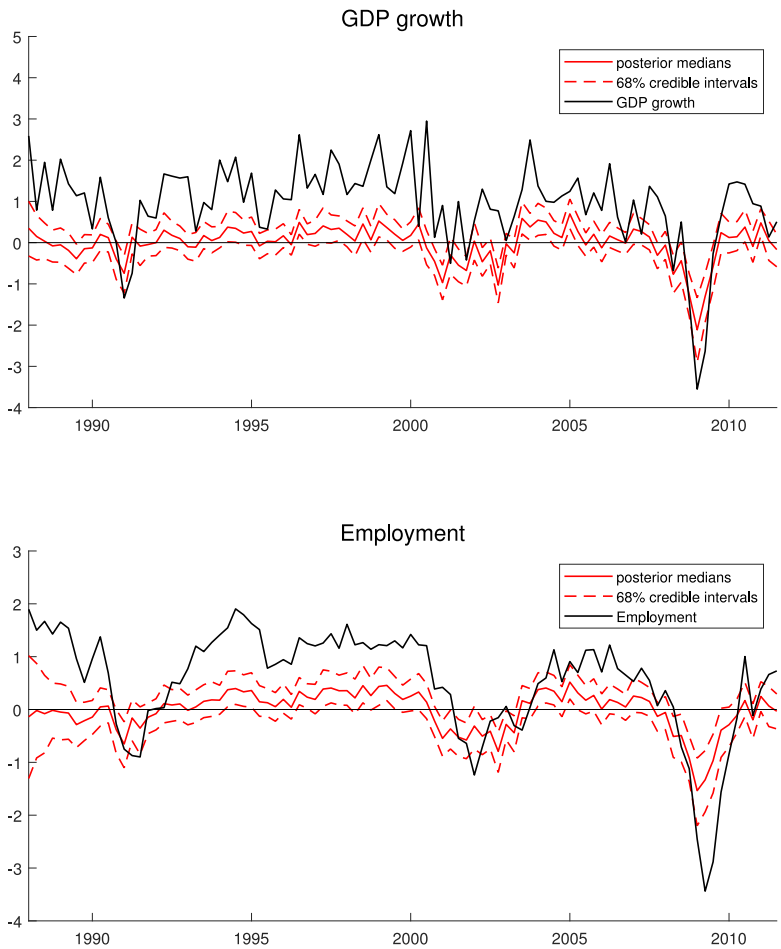

Fig. 1. Posterior estimates of the joint historical contributions of the oil, monetary policy, productivity, liquidity and financial risk, and fiscal policy shocks to GDP growth and employment during 1987Q1-2011Q2.

4.3. Historical and forecast error variance decompositions

We estimate historical decompositions to assess how much variations in GDP growth and employment are contributed by the cumulative effects of oil, monetary policy, productivity, liquidity and financial risk, and fiscal policy shocks.

Fig. 1 presents the posterior medians and their corresponding 68% credible intervals for the joint contributions of the 5 structural shocks to GDP growth and employment over the period from 1987Q1 to 2011Q2. Our results indicate that these 5 structural shocks jointly explain only a moderate fraction of the variations in both GDP growth and employment during the Great Moderation period in the 1990s. However, starting from 2000 onwards, the aggregate effects of these 5 structural shocks have become increasingly important and contribute to a considerable portion of the decline in GDP growth and employment during the 2007–09 recession period.

To further investigate the role played by each of the 5 structural shocks, we display in Figs. 2 and 3 the posterior medians of the historical contributions of each shock to GDP growth and employment. Our results show that the liquidity and financial risk shocks have played a more important role than the other shocks in driving the historical fluctuations in GDP growth and employment, particularly after the early 2000s. In contrast, the contributions of the oil, monetary policy, and fiscal policy shocks are relatively small. While the productivity shocks account for a notable portion of the fluctuations in GDP growth, their impact on the variations in employment is comparatively modest.

To take a closer look at the contributions of each structural shock to the 2007–09 recession, Table 3 summarizes the posterior median estimates of the historical contributions, as reported in Figs. 2 and 3, for GDP growth and employment at 2009Q2. It is immediately clear that the liquidity and financial risk shocks are the primary drivers of the 2007–09 recession, consistent with the findings in Stock and Watson (2012). Our results indicate that the liquidity and financial risk shocks account for about 50% of the decline in GDP growth and 30% of the decline in employment. The Productivity shocks emerge as the second most important among the 5 structural shocks, explaining about 9% of the decline in GDP growth and 4% of the decline in employment. The impacts of

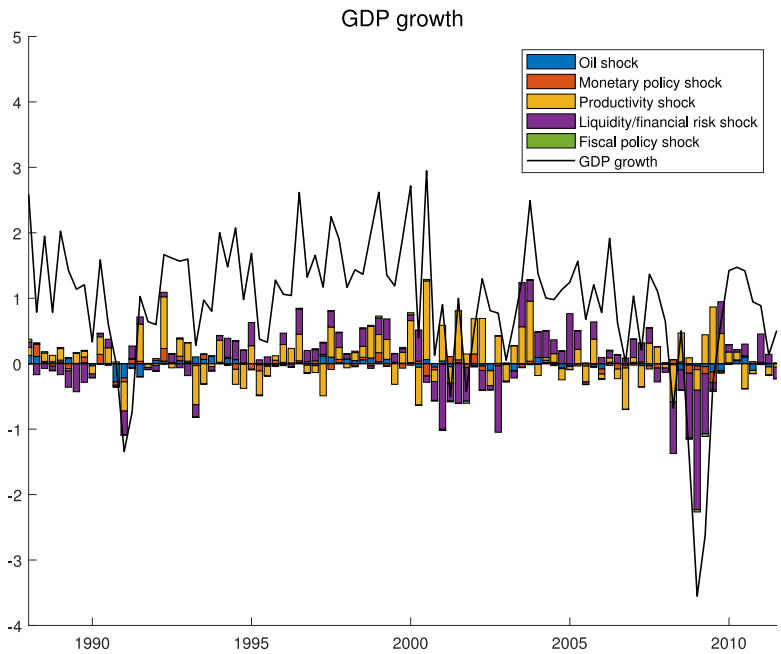

Fig. 2. Posterior medians of the historical contributions of the oil, monetary policy, productivity, liquidity and financial risk, and fiscal policy shocks to GDP growth during 1987Q1-2011Q2.

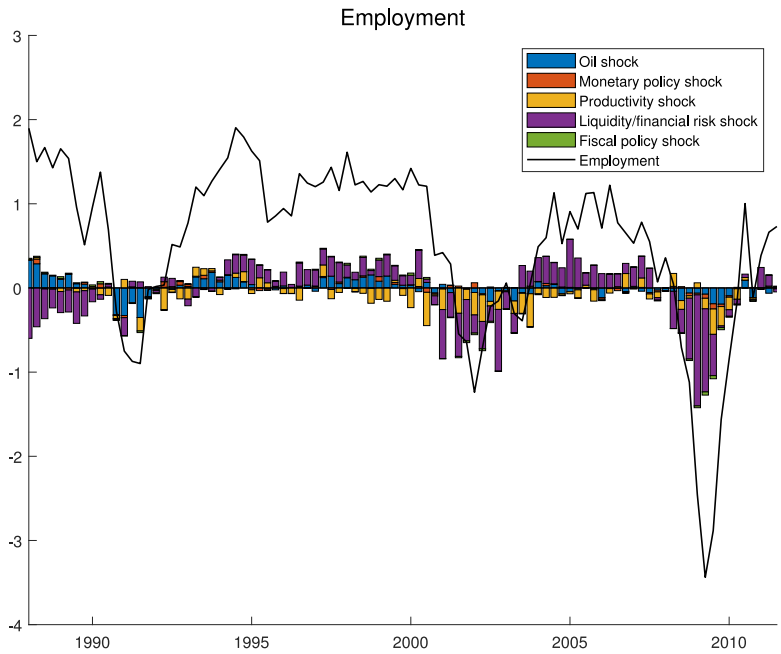

Fig. 3. Posterior medians of the historical contributions of the oil, monetary policy, productivity, liquidity and financial risk, and fiscal policy shocks to employment during 1987Q1-2011Q2.

the oil, monetary policy, and fiscal policy shocks are relatively small, with none of these shocks contributing more than 3% to the declines in both GDP growth and employment.

Table 4 presents the posterior medians of the forecast error variance decompositions for GDP growth and employment at both short horizon $H = 1$ and long horizon $H = 24$. Our results reveal that the 5 structural shocks collectively account for about 16% of the variation in GDP growth at $H = 1$ and 14% at $H = 24$. Notably, among these 5 structural shocks, the liquidity and financial risk shocks play the most important role in explaining the forecast error variance of GDP growth, contributing over 6% of the variation

Table 3
Posterior medians of the historical contributions of the oil, monetary policy, productivity, liquidity and financial risk, and fiscal policy shocks to GDP growth and employment at 2009Q2.

	GDP growth	Employment
Oil	−1.0%	−2.3%
Monetary policy	−1.6%	−1.5%
Productivity	−8.9%	−3.6%
Liquidity and financial risk	−51.3%	−28.7%
Fiscal policy	−1.1%	−1.2%
Total	−63.9%	−37.3%

Table 4
Posterior medians of the forecast error variance decompositions of GDP growth and employment attributed to the oil, monetary policy, productivity, liquidity and financial risk, and fiscal policy shocks at horizons $H = 1$ and $H = 24$.

Shocks	GDP growth		Employment	
	$H = 1$	$H = 24$	$H = 1$	$H = 24$
Oil	3.9%	3.5%	4.0%	4.6%
Monetary policy	0.5%	0.5%	0.2%	0.3%
Productivity	2.1%	2.0%	0.2%	0.3%
Liquidity and financial risk	6.5%	5.6%	2.3%	2.5%
Fiscal policy	3.1%	2.8%	1.5%	1.7%
Total	16.1%	14.4%	8.2%	9.4%

at $H = 1$ and 5% at $H = 24$. For employment, our findings suggest that all 5 structural shocks jointly contribute only a small portion to the forecast error variances. For instance, the overall contribution of the 5 structural shocks to the forecast error variance of employment is around 8% at $H = 1$ and 9% at $H = 24$, and none of these structural shocks individually accounts for more than 5% of the forecast error variance of employment.

5. Conclusion

This paper introduces an MCMC algorithm for Bayesian inference in SVARs with linear restrictions. Our proposed algorithm is built upon a novel parameter transformation scheme that can be applied to estimate large SVARs with flexible forms of linear equality and inequality restrictions and handle over-identifying restrictions. In the application, we demonstrate the usefulness of the proposed methodology by using a large Bayesian Proxy-SVAR with multiple proxy variables to simultaneously identify 5 macroeconomic shocks — oil, monetary policy, productivity, liquidity and financial risk, and fiscal policy shocks, and examine their contributions to the 2007–09 recession. Our results indicate that liquidity and financial risk shocks have played the most important role in driving the 2007–09 recession, accounting for about 50% of the decline in GDP growth and 30% of the decline in employment, while the other shocks have relatively minor effects.

Appendix A

A.1. Conditional priors

The linear equality and inequality restrictions specified in (2) and (3) for $\mathbf{b}_1$, i.e., $\mathbf{R}_1^E\mathbf{b}_1 = \mathbf{d}_1$ and $\mathbf{l}_1 < \mathbf{R}_1^I\mathbf{b}_1 < \mathbf{u}_1$, imply that the transformed parameters $\mathbf{w}$ must satisfy the following linear equality and inequality restrictions:

$$\begin{aligned} \mathbf{R}_{\mathbf{w}}^E\mathbf{w} &= \mathbf{d}_1, \\ \mathbf{l}_1 < \mathbf{R}_{\mathbf{w}}^I\mathbf{w} &< \mathbf{u}_1, \end{aligned}$$

where $\mathbf{R}_{\mathbf{w}}^E = \mathbf{R}_1^E\mathbf{T}^{-1}$ and $\mathbf{R}_{\mathbf{w}}^I = \mathbf{R}_1^I\mathbf{T}^{-1}$. Since $\mathbf{w} = \mathbf{T}\mathbf{b}_1$ and $\mathbf{b}_1$ has a truncated Gaussian prior (4), it follows that the induced prior for $\mathbf{w} = (w_1, \mathbf{w}_{-1}')'$ is also truncated Gaussian:

$$\mathbf{w} \sim \mathcal{N}(\boldsymbol{\mu}, \boldsymbol{\Lambda}^{-1})\mathbf{1}(\mathbf{R}_{\mathbf{w}}^E\mathbf{w} = \mathbf{d}_1)\mathbf{1}(\mathbf{l}_1 < \mathbf{R}_{\mathbf{w}}^I\mathbf{w} < \mathbf{u}_1) \tag{35}$$

with $\boldsymbol{\mu} = \mathbf{T}\boldsymbol{\beta}_1$, $\boldsymbol{\Lambda} = \mathbf{T}'^{-1}\mathbf{K}_1\mathbf{T}^{-1}$. Partition $\boldsymbol{\mu}$ and $\boldsymbol{\Lambda}$ as

$$\boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \boldsymbol{\mu}_{-1} \end{pmatrix}, \quad \boldsymbol{\Lambda} = \begin{pmatrix} \lambda_{11} & \lambda_{21}' \\ \lambda_{21} & \boldsymbol{\Lambda}_{22} \end{pmatrix},$$

where μ_1 and λ_{11} are scalars, $\boldsymbol{\mu}_{-1}$ and λ_{21} are $(n-1)\times 1$ vectors, and $\boldsymbol{\Lambda}_{22}$ is a $(n-1)\times(n-1)$ matrix. Using a standard property of the Gaussian distribution, we obtain the following conditional priors for w_1 and $\mathbf{w}_{-1}$:

$$(\mathbf{w}_{-1}|w_1) \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{w}_{-1}}, \boldsymbol{\Lambda}_{\mathbf{w}_{-1}}^{-1})\mathbf{1}(\mathbf{R}_{\mathbf{w}}^E\mathbf{w} = \mathbf{d}_1)\mathbf{1}(\mathbf{l}_1 < \mathbf{R}_{\mathbf{w}}^I\mathbf{w} < \mathbf{u}_1), \tag{36}$$

$$(w_1|\mathbf{w}_{-1}) \sim \mathcal{N}(\mu_{w_1}, \lambda_{w_1}^{-1})\mathbf{1}(\mathbf{R}_{\mathbf{w}}^E\mathbf{w} = \mathbf{d}_1)\mathbf{1}(\mathbf{l}_1 < \mathbf{R}_{\mathbf{w}}^I\mathbf{w} < \mathbf{u}_1), \tag{37}$$

where $\boldsymbol{\mu}_{\mathbf{w}_{-1}} = \boldsymbol{\mu}_{-1} - \boldsymbol{\Lambda}_{22}^{-1}\lambda_{21}(w_1 - \mu_1)$, $\boldsymbol{\Lambda}_{\mathbf{w}_{-1}} = \boldsymbol{\Lambda}_{22}$, $\mu_{w_1} = \mu_1 - \lambda_{11}^{-1}\lambda_{21}'(\mathbf{w}_{-1} - \boldsymbol{\mu}_{-1})$ and $\lambda_{w_1} = \lambda_{11}$.

When there are no restrictions imposed on $\mathbf{b}_1$, as is the case we considered in Section 2.3, the conditional priors specified in (36) and (37) reduce to

$$(\mathbf{w}_{-1}|w_1) \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{w}_{-1}}, \boldsymbol{\Lambda}_{\mathbf{w}_{-1}}^{-1}),$$
$$(w_1|\mathbf{w}_{-1}) \sim \mathcal{N}(\mu_{w_1}, \lambda_{w_1}^{-1}),$$

which gives the conditional priors (11) and (12) in the main text of this paper.

Next, we will derive the conditional priors of w_1 and $\mathbf{w}_{-1,1}$ when linear restrictions are imposed on $\mathbf{b}_1$, which is the case we consider in Section 2.4. Recall that the transformed parameters $\mathbf{w} = (w_1, \mathbf{w}_{-1})$ obtained from (15) can be written as $\mathbf{w} = (w_1, \mathbf{w}_{-1,1}', \mathbf{w}_{-1,2}')'$ with $\mathbf{w}_{-1} = (\mathbf{w}_{-1,1}', \mathbf{w}_{-1,2}')'$, $\mathbf{w}_{-1,1} = (w_2, \dots, w_{n-r_1^E})'$ and $\mathbf{w}_{-1,2} = (w_{n-r_1^E+1}, \dots, w_n)' = \mathbf{d}_1$. Thus, the conditional prior (36) for $\mathbf{w}_{-1} = (\mathbf{w}_{-1,1}', \mathbf{w}_{-1,2}')'$ becomes

$$(\mathbf{w}_{-1}|w_1) \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{w}_{-1}}, \boldsymbol{\Lambda}_{\mathbf{w}_{-1}}^{-1})\mathbf{1}(\mathbf{w}_{-1,2} = \mathbf{d}_1)\mathbf{1}(\mathbf{l}_1 < \mathbf{R}_{\mathbf{w}}^I\mathbf{w} < \mathbf{u}_1). \tag{38}$$

Note that $\mathcal{N}(\boldsymbol{\mu}_{\mathbf{w}_{-1}}, \boldsymbol{\Lambda}_{\mathbf{w}_{-1}}^{-1})\mathbf{1}(\mathbf{w}_{-1,2} = \mathbf{d}_1)$ is just a conditional Gaussian distribution with its last r_1^E elements, i.e., $\mathbf{w}_{-1,2}$, fixed at $\mathbf{d}_1$.[12] Partition $\boldsymbol{\mu}_{\mathbf{w}_{-1}}$ and $\boldsymbol{\Lambda}_{\mathbf{w}_{-1}}$ as

$$\boldsymbol{\mu}_{\mathbf{w}_{-1}} = \begin{pmatrix} \widetilde{\boldsymbol{\mu}}_1 \\ \widetilde{\boldsymbol{\mu}}_2 \end{pmatrix}, \quad \boldsymbol{\Lambda}_{\mathbf{w}_{-1}} = \begin{pmatrix} \widetilde{\boldsymbol{\Lambda}}_{11} & \widetilde{\boldsymbol{\Lambda}}_{21}' \\ \widetilde{\boldsymbol{\Lambda}}_{21} & \widetilde{\boldsymbol{\Lambda}}_{22} \end{pmatrix},$$

where $\widetilde{\boldsymbol{\mu}}_1$ is $(n - r_1^E - 1) \times 1$, $\widetilde{\boldsymbol{\mu}}_2$ is $r_1^E \times 1$, $\widetilde{\boldsymbol{\Lambda}}_{11}$ is $(n - r_1^E - 1) \times (n - r_1^E - 1)$, $\widetilde{\boldsymbol{\Lambda}}_{21}$ is $r_1^E \times (n - r_1^E - 1)$ and $\widetilde{\boldsymbol{\Lambda}}_{22}$ is $r_1^E \times r_1^E$. Then we obtain the conditional prior for $\mathbf{w}_{-1,1}$:

$$(\mathbf{w}_{-1,1}|w_1, \mathbf{w}_{-1,2}) \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{w}_{-1,1}}, \boldsymbol{\Lambda}_{\mathbf{w}_{-1,1}}^{-1})\mathbf{1}(\mathbf{l}_1 < \mathbf{R}_{\mathbf{w}}^I\mathbf{w} < \mathbf{u}_1) \tag{39}$$

with $\boldsymbol{\mu}_{\mathbf{w}_{-1,1}} = \widetilde{\boldsymbol{\mu}}_1 - \widetilde{\boldsymbol{\Lambda}}_{11}^{-1}\widetilde{\boldsymbol{\Lambda}}_{21}'(\mathbf{d}_1 - \widetilde{\boldsymbol{\mu}}_2)$ and $\boldsymbol{\Lambda}_{\mathbf{w}_{-1,1}} = \widetilde{\boldsymbol{\Lambda}}_{11}$, and the conditional prior for w_1 is given by

$$(w_1|\mathbf{w}_{-1}) \sim \mathcal{N}(\mu_{w_1}, \lambda_{w_1}^{-1})\mathbf{1}(\mathbf{l}_1 < \mathbf{R}_{\mathbf{w}}^I\mathbf{w} < \mathbf{u}_1). \tag{40}$$

The expressions (39) and (40) can be further simplified. To see this, we first partition $\mathbf{R}_{\mathbf{w}}^I$ as

$$\mathbf{R}_{\mathbf{w}}^I = \left(\mathbf{R}_{w_1}^I, \mathbf{R}_{\mathbf{w}_{-1,1}}^I, \mathbf{R}_{\mathbf{w}_{-1,2}}^I\right),$$

where $\mathbf{R}_{w_1}^I$ is $r_1^I \times 1$, $\mathbf{R}_{\mathbf{w}}^I$ is $r_1^I \times (n - r_1^E - 1)$ and $\mathbf{R}_{\mathbf{w}_{-1,1}}^I$ is $r_1^I \times r_1^E$. Then we can equivalently represent (39) and (40) as

$$(\mathbf{w}_{-1,1}|w_1, \mathbf{w}_{-1,2}) \sim \mathcal{N}(\boldsymbol{\mu}_{\mathbf{w}_{-1,1}}, \boldsymbol{\Lambda}_{\mathbf{w}_{-1,1}}^{-1})\mathbf{1}(\mathbf{l}_{\mathbf{w}_{-1,1}} < \mathbf{R}_{\mathbf{w}_{-1,1}}^I\mathbf{w}_{-1,1} < \mathbf{u}_{\mathbf{w}_{-1,1}}),$$
$$(w_1|\mathbf{w}_{-1}) \sim \mathcal{N}(\mu_{w_1}, \lambda_{w_1}^{-1})\mathbf{1}(\mathbf{l}_{w_1} < \mathbf{R}_{w_1}^I w_1 < \mathbf{u}_{w_1}),$$

where

$$\mathbf{l}_{\mathbf{w}_{-1,1}} = \mathbf{l}_1 - \mathbf{R}_{\mathbf{w}_{-1,2}}^I\mathbf{d}_1 - \mathbf{R}_{w_1}^I w_1,$$
$$\mathbf{u}_{\mathbf{w}_{-1,1}} = \mathbf{u}_1 - \mathbf{R}_{\mathbf{w}_{-1,2}}^I\mathbf{d}_1 - \mathbf{R}_{w_1}^I w_1,$$
$$\mathbf{l}_{w_1} = \mathbf{l}_1 - \mathbf{R}_{\mathbf{w}_{-1,2}}^I\mathbf{d}_1 - \mathbf{R}_{\mathbf{w}_{-1,1}}^I\mathbf{w}_{-1,1},$$
$$\mathbf{u}_{w_1} = \mathbf{u}_1 - \mathbf{R}_{\mathbf{w}_{-1,2}}^I\mathbf{d}_1 - \mathbf{R}_{\mathbf{w}_{-1,1}}^I\mathbf{w}_{-1,1}.$$

Note that, by construction, $(\mathbf{R}_{\mathbf{w}_{-1,1}}^I, \mathbf{l}_{\mathbf{w}_{-1,1}}, \mathbf{u}_{\mathbf{w}_{-1,1}})$ does not depend on $\mathbf{w}_{-1,1}$, and $(\mathbf{R}_{w_1}^I, \mathbf{l}_{w_1}, \mathbf{u}_{w_1})$ does not depend on w_1. This gives the conditional priors (16) and (17) in the main text of this paper.

A.2. Proof of Lemma 1

Given the transformation $\mathbf{T}$ specified in (5), by a direct computation, the transformed parameter matrix $\widetilde{\mathbf{B}} = \mathbf{TB}$ is of the form:

$$\widetilde{\mathbf{B}} = \begin{pmatrix} w_1 & \mathbf{0}_{1\times(n-1)} \\ \mathbf{w}_{-1} & \widetilde{\mathbf{B}}_{22} \end{pmatrix},$$

where $\widetilde{\mathbf{B}}_{22}$ is a $(n-1) \times (n-1)$ matrix. Thus, we have

$$\widetilde{\mathbf{B}}^{-1} = \begin{pmatrix} w_1^{-1} & \mathbf{0}_{1\times(n-1)} \\ -w_1^{-1}\widetilde{\mathbf{B}}_{22}^{-1}\mathbf{w}_{-1} & \widetilde{\mathbf{B}}_{22}^{-1} \end{pmatrix}, \tag{41}$$

[12] The implied restrictions $\mathbf{w}_{-1,2} = \mathbf{d}_1$ in (38) can also be directly verified from the linear restrictions $\mathbf{R}_{\mathbf{w}}^E\mathbf{w} = \mathbf{d}_1$ in the conditional prior (36), by noting that, with $\mathbf{T}$ given in (15), $\mathbf{R}_{\mathbf{w}}^E = \mathbf{R}_1^E\mathbf{T}^{-1} = (\mathbf{0}_{r_1^E\times(n-r_1^E)}, \mathbf{I}_{r_1^E})$. Therefore, it follows that $\mathbf{R}_{\mathbf{w}}^E\mathbf{w} = \mathbf{w}_{-1,2} = \mathbf{d}_1$.

$$\det(\widetilde{\mathbf{B}}) = w_1 \det(\widetilde{\mathbf{B}}_{22}). \tag{42}$$

It is known that the expression (41) holds only when $\widetilde{\mathbf{B}}_{22}$ is invertible. However, since $\widetilde{\mathbf{B}}$ is invertible, (42) implies that both w_1 and $\det(\widetilde{\mathbf{B}}_{22})$ are nonzero, which in turn ensures that $\widetilde{\mathbf{B}}_{22}$ is invertible.

Let $\mathbf{C_D}$ denote the Cholesky factor of $\mathbf{D} = \widetilde{\mathbf{Y}}'\widetilde{\mathbf{Y}} \otimes \mathbf{I}_n$ such that $\mathbf{D} = \mathbf{C}_\mathbf{D}'\mathbf{C_D}$. From the expression (41), we can write $\mathbf{C_D}\mathrm{vec}(\widetilde{\mathbf{B}}^{-1})$ as

$$\mathbf{C_D}\mathrm{vec}(\widetilde{\mathbf{B}}^{-1}) = \mathbf{C_D}\begin{pmatrix} w_1^{-1} \\ -w_1^{-1}\widetilde{\mathbf{B}}_{22}^{-1}\mathbf{w}_{-1} \\ \mathrm{vec}\begin{pmatrix} \mathbf{0}_{1\times(n-1)} \\ \widetilde{\mathbf{B}}_{22}^{-1} \end{pmatrix} \end{pmatrix}.$$

Here, we provide two expressions for $\mathbf{C_D}\mathrm{vec}(\widetilde{\mathbf{B}}^{-1})$:

$$\mathbf{C_D}\mathrm{vec}(\widetilde{\mathbf{B}}^{-1}) = \mathbf{q}_{\mathbf{w}_{-1}} - \mathbf{Z}_{\mathbf{w}_{-1}}\mathbf{w}_{-1}, \tag{43}$$

$$\mathbf{C_D}\mathrm{vec}(\widetilde{\mathbf{B}}^{-1}) = \mathbf{q}_{w_1} + w_1^{-1}\mathbf{Z}_{w_1}, \tag{44}$$

where

$$\mathbf{q}_{\mathbf{w}_{-1}} = \mathbf{C_D}\begin{pmatrix} w_1^{-1} \\ \mathbf{0}_{(n-1)\times 1} \\ \mathrm{vec}\begin{pmatrix} \mathbf{0}_{1\times(n-1)} \\ \widetilde{\mathbf{B}}_{22}^{-1} \end{pmatrix} \end{pmatrix}, \quad \mathbf{Z}_{\mathbf{w}_{-1}} = \mathbf{C_D}\begin{pmatrix} \mathbf{0}_{1\times(n-1)} \\ w_1^{-1}\widetilde{\mathbf{B}}_{22}^{-1} \\ \mathbf{0}_{n(n-1)\times(n-1)} \end{pmatrix},$$

$$\mathbf{q}_{w_1} = \mathbf{C_D}\begin{pmatrix} \mathbf{0}_{n\times 1} \\ \mathrm{vec}\begin{pmatrix} \mathbf{0}_{1\times(n-1)} \\ \widetilde{\mathbf{B}}_{22}^{-1} \end{pmatrix} \end{pmatrix}, \quad \mathbf{Z}_{w_1} = \mathbf{C_D}\begin{pmatrix} 1 \\ -\widetilde{\mathbf{B}}_{22}^{-1}\mathbf{w}_{-1} \\ \mathbf{0}_{n(n-1)\times 1} \end{pmatrix}.$$

Notice that $(\mathbf{q}_{\mathbf{w}_{-1}}, \mathbf{Z}_{\mathbf{w}_{-1}})$ does not depend on $\mathbf{w}_{-1}$ and $(\mathbf{q}_{w_1}, \mathbf{Z}_{w_1})$ does not depend on w_1. Hence, by substituting (42) and (43) into the likelihood function (8), we obtain the conditional likelihood of $\mathbf{w}_{-1}$:

$$\begin{aligned} p_{\mathbf{w}_{-1}}(\widetilde{\mathbf{y}}|w_1, \mathbf{w}_{-1}, \mathbf{B}_{-1}) &\propto |\det(\widetilde{\mathbf{B}})|^{-T} e^{-\frac{1}{2}\mathrm{vec}(\widetilde{\mathbf{B}}^{-1})'\mathbf{D}\mathrm{vec}(\widetilde{\mathbf{B}}^{-1})} \\ &\propto |w_1 \det(\widetilde{\mathbf{B}}_{22})|^{-T} e^{-\frac{1}{2}\mathrm{vec}(\widetilde{\mathbf{B}}^{-1})'\mathbf{C}_\mathbf{D}'\mathbf{C_D}\mathrm{vec}(\widetilde{\mathbf{B}}^{-1})} \\ &\propto e^{-\frac{1}{2}(\mathbf{q}_{\mathbf{w}_{-1}} - \mathbf{Z}_{\mathbf{w}_{-1}}\mathbf{w}_{-1})'(\mathbf{q}_{\mathbf{w}_{-1}} - \mathbf{Z}_{\mathbf{w}_{-1}}\mathbf{w}_{-1})}. \end{aligned}$$

By substituting (42) and (44) into the likelihood function (8), we obtain the conditional likelihood of w_1:

$$\begin{aligned} p_{w_1}(\widetilde{\mathbf{y}}|w_1, \mathbf{w}_{-1}, \mathbf{B}_{-1}) &\propto |\det(\widetilde{\mathbf{B}})|^{-T} e^{-\frac{1}{2}\mathrm{vec}(\widetilde{\mathbf{B}}^{-1})'\mathbf{D}\mathrm{vec}(\widetilde{\mathbf{B}}^{-1})} \\ &\propto |w_1 \det(\widetilde{\mathbf{B}}_{22})|^{-T} e^{-\frac{1}{2}\mathrm{vec}(\widetilde{\mathbf{B}}^{-1})'\mathbf{C}_\mathbf{D}'\mathbf{C_D}\mathrm{vec}(\widetilde{\mathbf{B}}^{-1})} \\ &\propto |w_1|^{-T} e^{-\frac{1}{2}(\mathbf{q}_{w_1} - w_1^{-1}\mathbf{Z}_{w_1})'(\mathbf{q}_{w_1} - w_1^{-1}\mathbf{Z}_{w_1})}, \\ &\propto |w_1|^{-T} e^{-\frac{1}{2}(w_1^{-1}\gamma_1 + w_1^{-2}\gamma_2)} \end{aligned}$$

where

$$\gamma_1 = 2\mathbf{Z}_{w_1}'\mathbf{q}_{w_1}, \quad \gamma_2 = \mathbf{Z}_{w_1}'\mathbf{Z}_{w_1}.$$

Therefore, we complete the proof of Lemma 1.

A.3. Proof of Proposition 2

Proposition 2 follows from the standard change of variable result. To be specific, we have

$$q^*(w_1^*) = \widetilde{q}(\widetilde{w}_1)\left|\frac{d\widetilde{w}_1}{dw_1^*}\right|,$$

where $q^*(w_1^*)$ denotes the density function of w_1^* and $\widetilde{q}(\widetilde{w}_1)$ denotes the density function of $\widetilde{w}_1$. As the inverse mapping implies $\widetilde{w}_1 = w_1^{*-1}\sqrt{\frac{\gamma_2}{T-2}}$, then

$$\left|\frac{d\widetilde{w}_1}{dw_1^*}\right| = \left|-w_1^{*-2}\sqrt{\frac{\gamma_2}{T-2}}\right| \propto |w_1^*|^{-2}. \tag{45}$$

Next, since $\widetilde{w}_1 \sim \mathcal{AN}\left(-\frac{\gamma_1}{2\sqrt{\gamma_2(T-2)}}, \frac{1}{T-2}\right)$, the density function of $\widetilde{w}_1$ is given by

$$\widetilde{q}(\widetilde{w}_1) \propto |\widetilde{w}_1|^{T-2} e^{-\frac{T-2}{2}\left(\widetilde{w}_1 + \frac{\gamma_1}{2\sqrt{\gamma_2(T-2)}}\right)^2}$$

$$\propto |\widetilde{w}_1|^{T-2} e^{-\frac{T-2}{2}\left(\widetilde{w}_1^2+\widetilde{w}_1\frac{\gamma_1}{\sqrt{\gamma_2(T-2)}}\right)}$$
$$\propto |\widetilde{w}_1|^{T-2} e^{-\frac{T-2}{2}\widetilde{w}_1^2-\frac{\gamma_1}{2}\widetilde{w}_1\sqrt{\frac{T-2}{\gamma_2}}}.$$

By substituting $\widetilde{w}_1 = w_1^{*-1}\sqrt{\frac{\gamma_2}{T-2}}$ into the above equation, we can express $\widetilde{q}(\widetilde{w}_1)$ in terms of w_1^* and obtain

$$\widetilde{q}(\widetilde{w}_1) \propto |w_1^*|^{-T+2} e^{-\frac{1}{2}(w_1^{*-1}\gamma_1+w_1^{*-2}\gamma_2)}. \tag{46}$$

Combining the results in (45) and (46), we obtain

$$q^*(w_1^*) = \widetilde{q}(\widetilde{w}_1)\left|\frac{d\widetilde{w}_1}{dw_1^*}\right| \propto |w_1^*|^{-T} e^{-\frac{1}{2}(w_1^{*-1}\gamma_1+w_1^{*-2}\gamma_2)}.$$

This completes the proof.

A.4. On the assumption $\mathrm{rk}[(\mathbf{B}_{-1}^{\perp}, \mathbf{R}_1^{E\prime})'] = r_1^E + 1$

In Section 2.4, we make an assumption that $\mathrm{rk}[(\mathbf{B}_{-1}^{\perp}, \mathbf{R}_1^{E\prime})'] = r_1^E + 1$. In this Appendix, we show that this assumption can be relaxed by appropriately reformulating the linear equality restrictions $\mathbf{R}_1^E\mathbf{b}_1 = \mathbf{d}_1$.

Since $\mathrm{rk}(\mathbf{R}_1^E) = r_1^E$, if $\mathrm{rk}[(\mathbf{B}_{-1}^{\perp}, \mathbf{R}_1^{E\prime})'] \neq r_1^E + 1$, then there is a $r_1^E \times 1$ vector $\boldsymbol{\tau}$ such that $\boldsymbol{\tau}'\mathbf{R}_1^E = \mathbf{B}_{-1}^{\perp\prime}$.[13] Let $\boldsymbol{\tau}^{\perp}$ be the orthogonal complement of $\boldsymbol{\tau}$ and define $\widehat{\mathbf{T}} = (\boldsymbol{\tau}, \boldsymbol{\tau}^{\perp})'$. As $\widehat{\mathbf{T}}$ is invertible, premultiplying $\widehat{\mathbf{T}}$ to $\mathbf{R}_1^E\mathbf{b}_1 = \mathbf{d}_1$, we have the following reformulated linear equality restrictions:

$$\widehat{\mathbf{R}}_1^E\mathbf{b}_1 = \widehat{\mathbf{d}}_1, \tag{47}$$

where $\widehat{\mathbf{R}}_1^E = \widehat{\mathbf{T}}\mathbf{R}_1^E$ and $\widehat{\mathbf{d}}_1 = \widehat{\mathbf{T}}\mathbf{d}_1 = (\widehat{d}_{1,1}, \dots, \widehat{d}_{1,r_1^E})'$. Let $\widehat{\mathbf{R}}_{1,-1}^E$ be the same as $\widehat{\mathbf{R}}_1^E$ but with its first row deleted. Then, by construction, we can write $\widehat{\mathbf{R}}_1^E = (\mathbf{B}_{-1}^{\perp}, \widehat{\mathbf{R}}_{1,-1}^{E\prime})'$. Therefore, (47) can be equivalently represented as

$$\mathbf{B}_{-1}^{\perp\prime}\mathbf{b}_1 = \widehat{d}_{1,1}, \tag{48}$$
$$\widehat{\mathbf{R}}_{1,-1}^E\mathbf{b}_1 = \widehat{\mathbf{d}}_{1,-1}, \tag{49}$$

where $\widehat{\mathbf{d}}_{1,-1} = (\widehat{d}_{1,2}, \dots, \widehat{d}_{1,r_1^E})'$. Let $\mathbf{T}_{-1}^*$ be a $(n - r_1^E - 2)\times n$ matrix whose rows and the rows of $\widehat{\mathbf{R}}_1^E$ are linearly independent. One can set $\mathbf{T}_{-1}^{*\prime}$ to be a orthogonal complement of $\widehat{\mathbf{R}}_1^{E\prime}$. Define $\mathbf{T}_{-1} = (\mathbf{T}_{-1}^{*\prime}, \widehat{\mathbf{R}}_1^{E\prime})'$, then the parameter transformation (14) becomes

$$\mathbf{T} = \begin{pmatrix} \mathbf{B}_{-1}^{\perp\prime} \\ \mathbf{T}_{-1}^* \\ \widehat{\mathbf{R}}_{1,-1}^E \end{pmatrix}. \tag{50}$$

Given (48), (49) and (50), we can show that the parameter transformation (15) can be represented as

$$\mathbf{T}\mathbf{b}_1 = \mathbf{w} = \begin{pmatrix} w_1 \\ \mathbf{w}_{-1,1} \\ \mathbf{w}_{-1,2} \end{pmatrix} = \begin{pmatrix} \widehat{d}_{1,1} \\ \mathbf{w}_{-1,1} \\ \widehat{\mathbf{d}}_{1,-1} \end{pmatrix}. \tag{51}$$

It is clear from (51) that, by reformulating the original linear equality restrictions $\mathbf{R}_1^E\mathbf{b}_1 = \mathbf{d}_1$ into (48) and (49), the transformed parameter vector $\mathbf{w} = (w_1, \mathbf{w}_{-1,1}', \mathbf{w}_{-1,2}')'$ is restricted so that $w_1 = \widehat{d}_{1,1}$ and $\mathbf{w}_{-1,2} = \widehat{\mathbf{d}}_{1,-1}$. This leaves $\mathbf{w}_{-1,1}$ as the only free parameter vector that needs to be sampled, which can be drawn by following the similar procedure as described in Section 2.4.

Appendix B. Proxy-SVAR with NGN prior

This Appendix discusses how our proposed algorithm can be used to estimate a Proxy-SVAR with the NGN prior. Let $\mathbf{y}_t = (\boldsymbol{\zeta}_t', \mathbf{m}_t')'$ where $\boldsymbol{\zeta}_t$ is a $\widetilde{n}\times 1$ vector of endogenous variables and $\mathbf{m}_t$ is a $k\times 1$ vector of proxy variables. We consider the following A-model type Proxy-SVAR:

$$\bar{\mathbf{B}}\mathbf{y}_t = \bar{\mathbf{c}} + \bar{\mathbf{A}}_1\mathbf{y}_{t-1} + \cdots + \bar{\mathbf{A}}_p\mathbf{y}_{t-p} + \bar{\boldsymbol{\epsilon}}_t, \quad \bar{\boldsymbol{\epsilon}}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_n), \tag{52}$$

where $\bar{\mathbf{c}}$ is an $n\times 1$ vector of intercepts, $\bar{\mathbf{B}}$ and $\bar{\mathbf{A}}_i$, for $i = 1, \dots, p$, are $n\times n$ coefficient matrices with $\bar{\mathbf{B}}$ being invertible. According to Arias et al. (2021), the model parameter matrices take the following form:

$$\bar{\mathbf{B}} = \begin{pmatrix} \bar{\boldsymbol{\Gamma}}_0 & \mathbf{0}_{\widetilde{n}\times k} \\ \bar{\boldsymbol{\Phi}}_0 & \bar{\boldsymbol{\Psi}}_0 \end{pmatrix}, \quad \bar{\mathbf{A}}_i = \begin{pmatrix} \bar{\boldsymbol{\Gamma}}_i & \mathbf{0}_{\widetilde{n}\times k} \\ \bar{\boldsymbol{\Phi}}_i & \bar{\boldsymbol{\Psi}}_i \end{pmatrix}, \quad i = 1. \dots, p. \tag{53}$$

[13] In fact $\boldsymbol{\tau}$ is the unique solution for the linear system $\boldsymbol{\tau}'\mathbf{R}_1^E = \mathbf{B}_{-1}^{\perp\prime}$, which is given by $\boldsymbol{\tau} = \left(\mathbf{R}_1^E\mathbf{R}_1^{E\prime}\right)^{-1}\mathbf{R}_1^E\mathbf{B}_{-1}^{\perp}$.

where $\bar{\boldsymbol{\Gamma}}_i$ is $\tilde{n}\times\tilde{n}$, $\bar{\boldsymbol{\Phi}}_i$ is $k\times\tilde{n}$ and $\bar{\boldsymbol{\Psi}}_i$ is $k\times k$, for $i=0,\dots,p$.

Note that the B-model type Proxy-SVAR, specified in (26), can be recovered by premultiplying (52) by $\bar{\mathbf{B}}^{-1}$ and setting $\mathbf{B}=\bar{\mathbf{B}}^{-1}$, $\mathbf{c}=\bar{\mathbf{B}}^{-1}\bar{\mathbf{c}}$, and $\mathbf{A}_i=\bar{\mathbf{B}}^{-1}\bar{\mathbf{A}}_i$ for $i=1,\dots,p$. Through direct computation, it can be verified that the parameter matrices $\mathbf{B}$ and $\mathbf{A}_i$ for $i=1,\dots,p$, have the same form as those given in (27). Moreover, we have

$$\boldsymbol{\Gamma}_0=\bar{\boldsymbol{\Gamma}}_0^{-1},\quad \boldsymbol{\Phi}_0=-\bar{\boldsymbol{\Psi}}_0^{-1}\bar{\boldsymbol{\Phi}}_0\bar{\boldsymbol{\Gamma}}_0^{-1},\quad \boldsymbol{\Psi}_0=\bar{\boldsymbol{\Psi}}_0^{-1},$$
$$\boldsymbol{\Gamma}_i=\bar{\boldsymbol{\Gamma}}_0^{-1}\bar{\boldsymbol{\Gamma}}_i,\quad \boldsymbol{\Phi}_i=-\bar{\boldsymbol{\Psi}}_0^{-1}\bar{\boldsymbol{\Phi}}_0\bar{\boldsymbol{\Gamma}}_0^{-1}\bar{\boldsymbol{\Gamma}}_i+\bar{\boldsymbol{\Psi}}_0^{-1}\bar{\boldsymbol{\Phi}}_i,\quad \boldsymbol{\Psi}_i=\bar{\boldsymbol{\Psi}}_0^{-1}\bar{\boldsymbol{\Psi}}_i,$$

for $i=1,\dots,p$. It means that the zero restrictions for the B-model type Proxy-SVAR given in (27) are automatically implied by the zero restrictions for the A-model type Proxy-SVAR given in (53), and vice versa.

We consider a NGN prior for the model parameters $(\bar{\mathbf{B}},\bar{\mathbf{A}}_+)$ conditional on the zero restrictions in (53) and the exogeneity condition (33) for $\bar{\mathbf{B}}^{-1}=\mathbf{B}$, which has the density:

$$p(\bar{\mathbf{B}},\bar{\mathbf{A}}_+)\propto|\det(\bar{\mathbf{B}})|^{\nu-n}e^{-\frac{1}{2}\mathrm{vec}(\bar{\mathbf{B}}')'\mathbf{G}\mathrm{vec}(\bar{\mathbf{B}}')}e^{-\frac{1}{2}\left(\mathrm{vec}(\bar{\mathbf{A}}_+')-\mathbf{F}\mathrm{vec}(\bar{\mathbf{B}}')\right)'\mathbf{H}^{-1}\left(\mathrm{vec}(\bar{\mathbf{A}}_+')-\mathbf{F}\mathrm{vec}(\bar{\mathbf{B}}')\right)}. \tag{54}$$

The NGN prior is characterized by the parameters $(\nu,\mathbf{F},\mathbf{G},\mathbf{H})$, where $\nu>n$, $\mathbf{G}$ is a $n^2\times n^2$ definite matrix, $\mathbf{H}$ is a $n(1+np)\times n(1+np)$ positive definite matrix, and $\mathbf{F}$ is a $n(1+np)\times n^2$ matrix.

Let $\bar{\mathbf{A}}_+=(\bar{\mathbf{c}},\bar{\mathbf{A}}_1,\dots,\bar{\mathbf{A}}_p)$, $\mathbf{Y}=(\mathbf{y}_1,\dots,\mathbf{y}_T)'$ and $\mathbf{X}=(\mathbf{X}_1,\dots,\mathbf{X}_T)'$ with $\mathbf{X}_t=(1,\mathbf{y}_{t-1}',\dots,\mathbf{y}_{t-p}')'$. Given the model specified in (52) and (54), as shown by Arias et al. (2021), the posterior distribution of $(\bar{\mathbf{B}},\bar{\mathbf{A}}_+)$ is also an NGN distribution:

$$p(\bar{\mathbf{B}},\bar{\mathbf{A}}_+|\mathbf{y})\propto|\det(\bar{\mathbf{B}})|^{\tilde{\nu}-n}e^{-\frac{1}{2}\mathrm{vec}(\bar{\mathbf{B}}')'\tilde{\mathbf{G}}\mathrm{vec}(\bar{\mathbf{B}}')}e^{-\frac{1}{2}\left(\mathrm{vec}(\bar{\mathbf{A}}_+')-\tilde{\mathbf{F}}\mathrm{vec}(\bar{\mathbf{B}}')\right)'\tilde{\mathbf{H}}^{-1}\left(\mathrm{vec}(\bar{\mathbf{A}}_+')-\tilde{\mathbf{F}}\mathrm{vec}(\bar{\mathbf{B}}')\right)},$$

where $\tilde{\nu}=T+\nu$, $\tilde{\mathbf{H}}=\left(\mathbf{I}_n\otimes\mathbf{X}'\mathbf{X}+\mathbf{H}^{-1}\right)^{-1}$, $\tilde{\mathbf{F}}=\tilde{\mathbf{H}}\left(\mathbf{I}_n\otimes\mathbf{X}'\mathbf{Y}+\mathbf{H}^{-1}\mathbf{F}\right)$ and $\tilde{\mathbf{G}}=\mathbf{I}_n\otimes\mathbf{Y}'\mathbf{Y}+\mathbf{G}+\mathbf{F}'\mathbf{H}^{-1}\mathbf{F}-\tilde{\mathbf{F}}'\tilde{\mathbf{H}}^{-1}\tilde{\mathbf{F}}$.

Posterior draws of $(\bar{\mathbf{B}},\bar{\mathbf{A}}_+)$ conditional on the required zero restrictions can be obtained via drawing from the following distributions:

$$p(\bar{\mathbf{B}}|\mathbf{y})\propto|\det(\bar{\mathbf{B}})|^{\tilde{\nu}-n}e^{-\frac{1}{2}\mathrm{vec}(\bar{\mathbf{B}}')'\tilde{\mathbf{G}}\mathrm{vec}(\bar{\mathbf{B}}')}, \tag{55}$$

$$p(\bar{\mathbf{A}}_+|\bar{\mathbf{B}},\mathbf{y})\propto e^{-\frac{1}{2}\left(\mathrm{vec}(\bar{\mathbf{A}}_+')-\tilde{\mathbf{F}}\mathrm{vec}(\bar{\mathbf{B}}')\right)'\tilde{\mathbf{H}}^{-1}\left(\mathrm{vec}(\bar{\mathbf{A}}_+')-\tilde{\mathbf{F}}\mathrm{vec}(\bar{\mathbf{B}}')\right)}. \tag{56}$$

This shows that the conditional posterior of $\bar{\mathbf{A}}_+$ is a Gaussian distribution conditional on the zero restrictions given in (53), which can be drawn from in a straightforward manner.

In what follows, we discuss how the techniques described in the main text of this paper can be used to draw $\bar{\mathbf{B}}$. First, applying a standard change of variable result to the posterior distribution (55) with the transformation $\mathbf{B}=\bar{\mathbf{B}}^{-1}$, we have

$$\begin{aligned}p(\mathbf{B}|\bar{\mathbf{A}}_+,\mathbf{y})&\propto|\det(\mathbf{B})|^{-(\tilde{\nu}+n)}e^{-\frac{1}{2}\mathrm{vec}(\mathbf{B}^{-1\prime})'\tilde{\mathbf{G}}\mathrm{vec}(\mathbf{B}^{-1\prime})}\\&\propto|\det(\mathbf{B})|^{-(\tilde{\nu}+n)}e^{-\frac{1}{2}\mathrm{vec}(\mathbf{B}^{-1})'\bar{\mathbf{G}}\mathrm{vec}(\mathbf{B}^{-1})},\end{aligned}$$

where $\bar{\mathbf{G}}=\mathbf{L}'\tilde{\mathbf{G}}\mathbf{L}$ and $\mathbf{L}$ is a commutation matrix such that $\mathbf{L}\mathrm{vec}(\mathbf{B}^{-1})=\mathrm{vec}(\mathbf{B}^{-1\prime})$. As we have discussed above, the zero restrictions on $\bar{\mathbf{B}}$ given in (53) implies that $\mathbf{B}$ is of the same form as specified in (27). Therefore, with the exogeneity condition, we need to draw from $p(\mathbf{B}|\bar{\mathbf{A}}_+,\mathbf{y})$ conditional on the zero restrictions that are the same as those given in (27) and (33). After obtaining draws of $\mathbf{B}$, we can recover the posterior draws of $\bar{\mathbf{B}}$ by the inverse transformation $\bar{\mathbf{B}}=\mathbf{B}^{-1}$.

Following the same line of the discussions in the main text of this paper, we can draw $\mathbf{B}$ using a Gibbs sampler that iteratively draws from the conditional posteriors $p(\mathbf{b}_i|\mathbf{B}_{-i},\mathbf{y})$ for $i=1,\dots,n$. This can be accomplished using our proposed method described in Section 2.4. In particular, when drawing from $p(\mathbf{b}_1|\mathbf{B}_{-1},\mathbf{y})$, we first define $\mathbf{w}=\mathbf{T}\mathbf{b}_1$, where $\mathbf{T}$ is constructed in a manner similar to that specified in (14), and $\mathbf{w}=(w_1,\mathbf{w}_{-1}')'=(w_1,\mathbf{w}_{-1,1}',\mathbf{w}_{-1,2})'$ is the first column of $\tilde{\mathbf{B}}=\mathbf{T}\mathbf{B}$. The zero restrictions on $\mathbf{b}_1$ imply that $\mathbf{w}_{-1,2}=\mathbf{0}$, which leaves $(w_1,\mathbf{w}_{-1,1})$ as the only free parameters to be sampled. Using a standard change of variable result with $\mathbf{w}=\mathbf{T}\mathbf{b}_1$, it can be shown that the conditional posterior of $(w_1,\mathbf{w}_{-1,1}')'$ is given by

$$p(w_1,\mathbf{w}_{-1,1}|\mathbf{w}_{-1,2},\mathbf{B}_{-1},\bar{\mathbf{A}}_+,\mathbf{y})\propto|\det(\tilde{\mathbf{B}})|^{-(\tilde{\nu}+n)}e^{-\frac{1}{2}\mathrm{vec}(\tilde{\mathbf{B}}^{-1})'\hat{\mathbf{G}}\mathrm{vec}(\tilde{\mathbf{B}}^{-1})}, \tag{57}$$

where $\hat{\mathbf{G}}=(\mathbf{T}\otimes\mathbf{I}_n)\bar{\mathbf{G}}(\mathbf{T}'\otimes\mathbf{I}_n)$. Note that this conditional posterior has the same form as the likelihood function given in (8). Hence, we can derive the conditional posteriors $p(\mathbf{w}_{-1}|w_1,\mathbf{B}_{-1},\bar{\mathbf{A}}_+,\mathbf{y})$ and $p(w_1|\mathbf{w}_{-1},\mathbf{B}_{-1},\bar{\mathbf{A}}_+,\mathbf{y})$ in the same way as we derive the conditional likelihoods $p_{\mathbf{w}_{-1}}(\tilde{\mathbf{y}}|w_1,\mathbf{w}_{-1},\mathbf{B}_{-1})$ and $p_{w_1}(\tilde{\mathbf{y}}|w_1,\mathbf{w}_{-1},\mathbf{B}_{-1})$ as described in Appendix A.2. It turns out that the conditional posterior of $\mathbf{w}_{-1,1}$ follows a Gaussian distribution, and the density of the conditional posterior of w_1 has the same form as (10), which can be drawn from using the method proposed in Proposition 2 or the exact sampling approach developed in Appendix C.

Appendix C. Exact sampling from $p(w_1|\mathbf{w}_{-1},\mathbf{B}_{-1},\tilde{\mathbf{y}})$

This Appendix describes an approach to achieve exact sampling from $p(w_1|\mathbf{w}_{-1},\mathbf{B}_{-1},\tilde{\mathbf{y}})$. To draw from $\mathcal{AN}(a,b)$, Villani (2009) proposes to use a normal mixture approximation:

$$f_{AN}(\tilde{w}_1;a,b)\approx p_{MN}(\tilde{w}_1),$$

where

$$p_{MN}(\widetilde{w}_1) = P\phi(\widetilde{w}_1; m_2, \sigma_1^2) + (1-P)\phi(\widetilde{w}_1; m_2, \sigma_2^2), \tag{58}$$

$\phi(x, \widetilde{m}, \widetilde{\sigma}^2)$ denotes the density of a normal distribution with mean $\widetilde{m}$ and variance $\widetilde{\sigma}^2$, $m_1 = \frac{a}{2} - \frac{1}{2}\sqrt{a^2+4}$, $m_2 = \frac{a}{2} + \frac{1}{2}\sqrt{a^2+4}$, $\sigma_i^2 = \frac{m_i^2}{1+m_i^2}b$, $i = 1, 2$, and $P = (1 + e^{\frac{2a}{b}})^{-1}$.

Suppose we draw $\widetilde{w}_1 \sim P\phi(\widetilde{w}_1; m_2, \sigma_1^2) + (1-P)\phi(\widetilde{w}_1; m_2, \sigma_2^2)$ and set $w_1^* = \widetilde{w}_1^{-1}\sqrt{\frac{\gamma_2}{T-2}}$, then using a standard change of variable result and (45) in Appendix A.3, it implies that the density function of w_1^* is given by

$$\widetilde{p}(w_1^*) \propto |w_1^*|^{-2}\left(P\phi\left(w_1^{*-1}\sqrt{\frac{\gamma_2}{T-1}}, \mu_1, \sigma_1^2\right) + (1-P)\phi\left(w_1^{*-1}\sqrt{\frac{\gamma_2}{T-1}}, \mu_1, \sigma_1^2\right)\right). \tag{59}$$

This shows that the method of normal mixture approximation proposed in Villani (2009) actually uses the density function (59) to approximate (13).

Next, we propose an independent Metropolis–Hastings step using the proposal distribution with density function (59) to obtain exact draws from the conditional posterior $p(w_1|\mathbf{w}_{-1}, \mathbf{B}_{-1}, \widetilde{\mathbf{y}}) \propto |w_1|^{-T} e^{-\frac{1}{2}(w_1^{-1}\gamma_1 + w_1^{-2}\gamma_2)} e^{-\frac{\lambda_{w_1}}{2}(w_1-\mu_{w_1})^2}$. To be specific, given a current draw w_1, when there are no restrictions imposed on the parameter space, we can first obtain a draw $\widetilde{w}_1$ from (58) and set $w_1^* = \widetilde{w}_1^{-1}\sqrt{\frac{\gamma_2}{T-2}}$, then the proposal w_1^* is accepted with probability

$$\alpha = \min\left\{1, \frac{|w_1^*|^{-T} e^{-\frac{1}{2}(w_1^{*-1}\gamma_1 + w_1^{*-2}\gamma_2)} e^{-\frac{\lambda_{w_1}}{2}(w_1^*-\mu_{w_1})^2}\widetilde{p}(w_1^*)}{|w_1|^{-T} e^{-\frac{1}{2}(w_1^{-1}\gamma_1 + w_1^{-2}\gamma_2)} e^{-\frac{\lambda_{w_1}}{2}(w_1-\mu_{w_1})^2}\widetilde{p}(w_1)}\right\}. \tag{60}$$

When the linear equality and inequality restrictions are imposed, then the acceptance probability becomes $\alpha \times \mathbf{1}(\mathbf{l}_{w_1} < \mathbf{R}_{w_1}^I w_1^* < \mathbf{u}_{w_1})$. The procedures for obtaining an exact draw from the conditional posterior $p(w_1|\mathbf{w}_{-1}, \mathbf{B}_{-1}, \widetilde{\mathbf{y}})$ are summarized in Algorithm 3, which can replace Step 3 in both Algorithm 1 and Algorithm 2.

Algorithm 3 Sampling $p(w_1|\mathbf{w}_{-1}, \mathbf{B}_{-1}, \widetilde{\mathbf{y}})$.

Given w_1, complete the following steps.

1. Set $a = -\frac{\gamma_1}{2\sqrt{\gamma_2(T-2)}}$ and $b = \frac{1}{T-2}$.
2. Draw $\widetilde{w}_1$ from $p_{MN}(\widetilde{w}_1)$, where $p_{MN}(\widetilde{w}_1)$ is defined in (58), and set $w_1^* = \widetilde{w}_1^{-1}\sqrt{\frac{\gamma_2}{T-2}}$.
3. Compute α specified as in (60), then
   (a) if there are no restrictions imposed on the parameter space, set $w_1 = w_1^*$ with probability α.
   (b) if the linear equality and inequality restrictions are imposed on the parameter space, set $w_1 = w_1^*$ with probability $\alpha \times \mathbf{1}(\mathbf{l}_{w_1} < \mathbf{R}_{w_1}^I w_1^* < \mathbf{u}_{w_1})$.

Appendix D. Supplementary material

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jeconom.2024.105850.

References

Amir-Ahmadi, P., Drautzburg, T., 2021. Identification and inference with ranking restrictions. Quant. Econ. 12 (1), 1–39.
Antolín-Díaz, J., Rubio-Ramírez, J.F., 2018. Narrative sign restrictions for SVARs. Amer. Econ. Rev. 108 (10), 2802–2829.
Arias, J.E., Rubio-Ramírez, J.F., Waggoner, D.F., 2018. Inference based on structural vector autoregressions identified with sign and zero restrictions: Theory and applications. Econometrica 86 (2), 685–720.
Arias, J.E., Rubio-Ramírez, J.F., Waggoner, D.F., 2021. Inference in Bayesian proxy-SVARs. J. Econometrics 225 (1), 88–106.
Arias, J., Rubio-Ramirez, J.F., Waggoner, D.F., 2023. Uniform Priors for Impulse Responses. FRB Atlanta Working Paper.
Baumeister, C., Hamilton, J.D., 2015. Sign restrictions, structural vector autoregressions, and useful prior information. Econometrica 83 (5), 1963–1999.
Baumeister, C., Hamilton, J.D., 2019. Structural interpretation of vector autoregressions with incomplete identification: Revisiting the role of oil supply and demand shocks. Amer. Econ. Rev. 109 (5), 1873–1910.
Bhattacharya, A., Pati, D., Pillai, N.S., Dunson, D.B., 2015. Dirichlet-Laplace priors for optimal shrinkage. J. Amer. Statist. Assoc. 110 (512), 1479–1490.
Botev, Z.I., 2017. The normal law under linear restrictions: Simulation and estimation via minimax tilting. J. R. Stat. Soc. Ser. B Stat. Methodol. 79 (1), 125–148.
Braun, R., Brüggemann, R., 2023. Identification of SVAR models by combining sign restrictions with external instruments. J. Bus. Econom. Statist. 41 (4), 1077–1089.
Brown, P.J., Griffin, J.E., 2010. Inference with normal-Gamma prior distributions in regression problems. Bayesian Anal. 5, 171–188.
Bruns, M., Piffer, M., 2023. A new posterior sampler for Bayesian structural vector autoregressive models. Quant. Econ. 14 (4), 1221–1250.
Caldara, D., Herbst, E., 2019. Monetary policy, real activity, and credit spreads: Evidence from Bayesian proxy SVARs. Am. Econ. J.: Macroecon. 11 (1), 157–192.
Carvalho, C.M., Polson, N.G., Scott, J.G., 2010. The horseshoe estimator for sparse signals. Biometrika 97 (2), 465–480.
Chan, J.C., 2021. Minnesota-type adaptive hierarchical priors for large Bayesian VARs. Int. J. Forecast. 37 (3), 1212–1226.
Chan, J., Eisenstat, E., Yu, X., 2022. Large Bayesian VARs with Factor Stochastic Volatility: Identification, Order Invariance and Structural Analysis. Working Paper.

Chan, J.C., Jeliazkov, I., 2009. Efficient simulation and integrated likelihood estimation in state space models. Int. J. Math. Model. Numer. Optim. 1 (1–2), 101–120.
Chan, J.C., Poon, A., Zhu, D., 2023. High-dimensional conditionally Gaussian state space models with missing data. J. Econometrics 236 (1), 105468.
Cross, J.L., Hou, C., Poon, A., 2020. Macroeconomic forecasting with large Bayesian VARs: Global-local priors and the illusion of sparsity. Int. J. Forecast. 36 (3), 899–915.
Follett, L., Yu, C., 2017. Achieving parsimony in Bayesian VARs with the horseshoe prior. arXiv preprint arXiv:1709.07524.
Gelfand, A.E., Dey, D.K., 1994. Bayesian model choice: Asymptotics and exact calculations. J. R. Stat. Soc. Ser. B Stat. Methodol. 56 (3), 501–514.
George, E.I., Sun, D., Ni, S., 2008. Bayesian stochastic search for VAR model restrictions. J. Econometrics 142 (1), 553–580.
Geweke, J., 1989. Bayesian inference in econometric models using Monte Carlo integration. Econometrica 57, 1317–1339.
Huber, F., Feldkircher, M., 2019. Adaptive shrinkage in Bayesian vector autoregressive models. J. Bus. Econom. Statist. 37 (1), 27–39.
Kilian, L., Lütkepohl, H., 2017. Structural Vector Autoregressive Analysis. Cambridge University Press.
Korobilis, D., 2022. A new algorithm for structural restrictions in Bayesian vector autoregressions. Eur. Econ. Rev. 148, 104241.
Lütkepohl, H., 2005. New Introduction to Multiple Time Series Analysis. Springer Science & Business Media.
McCausland, W.J., Miller, S., Pelletier, D., 2011. Simulation smoothing for state-space models: A computational efficiency analysis. Comput. Statist. Data Anal. 55 (1), 199–212.
Mertens, K., Montiel Olea, J.L., 2018. Marginal tax rates and income: New time series evidence. Q. J. Econ. 133 (4), 1803–1884.
Mertens, K., Ravn, M.O., 2013. The dynamic effects of personal and corporate income tax changes in the United States. Amer. Econ. Rev. 103 (4), 1212–1247.
Mertens, K., Ravn, M.O., 2014. A reconciliation of SVAR and narrative estimates of tax multipliers. J. Monetary Econ. 68, S1–S19.
Rodriguez-Yam, G., Davis, R.A., Scharf, L.L., 2004. Efficient gibbs sampling of truncated multivariate normal with application to constrained linear regression. Unpublished Manuscript.
Rubio-Ramirez, J.F., Waggoner, D.F., Zha, T., 2010. Structural vector autoregressions: Theory of identification and algorithms for inference. Rev. Econ. Stud. 77 (2), 665–696.
Rue, H., 2001. Fast sampling of Gaussian Markov random fields. J. R. Stat. Soc. Ser. B Stat. Methodol. 63 (2), 325–338.
Sims, C.A., 1980. Macroeconomics and reality. Econometrica 48, 1–48.
Stock, J., Watson, M., 2012. Disentangling the channels of the 2007–2009 recession. Brook. Pap. Econ. Act. 81–135.
Villani, M., 2009. Steady-state priors for vector autoregressions. J. Appl. Econometrics 24 (4), 630–650.
Waggoner, D.F., Zha, T., 2003. A gibbs sampler for structural vector autoregressions. J. Econom. Dynam. Control 28 (2), 349–366.